AI Client SRS (Software Requirements Specification): OpenAI-Compatible AI Service Provisioning

Document Status: Draft
Version: 0.16
Last Updated: 2026-02-26
Target Audience: Engineering, DevOps, QA, Product
Reference Spec: OpenAI API Reference | OpenAI Concepts | Model Context Protocol

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1. Executive Summary

Context

We are an enterprise software vendor providing development tools to enterprise customers. Our products—VS Code extensions, Eclipse plugins, and other IDE integrations—are installed on **developer workstations and laptops** (Windows, macOS, Linux). We are incorporating Generative AI features into these tools and need to provide flexibility for customers to configure any OpenAI API-compliant cloud service while maintaining enterprise-grade security and administrative control.

Strategic Rationale

A key driver for this architecture is **reducing friction in the evaluation phase of the sales cycle**. Prospective customers evaluating our AI-powered tools should not be required to stand up AI gateways, deploy proxy infrastructure, or configure enterprise middleware before they can experience the product's value. The only prerequisites should be:

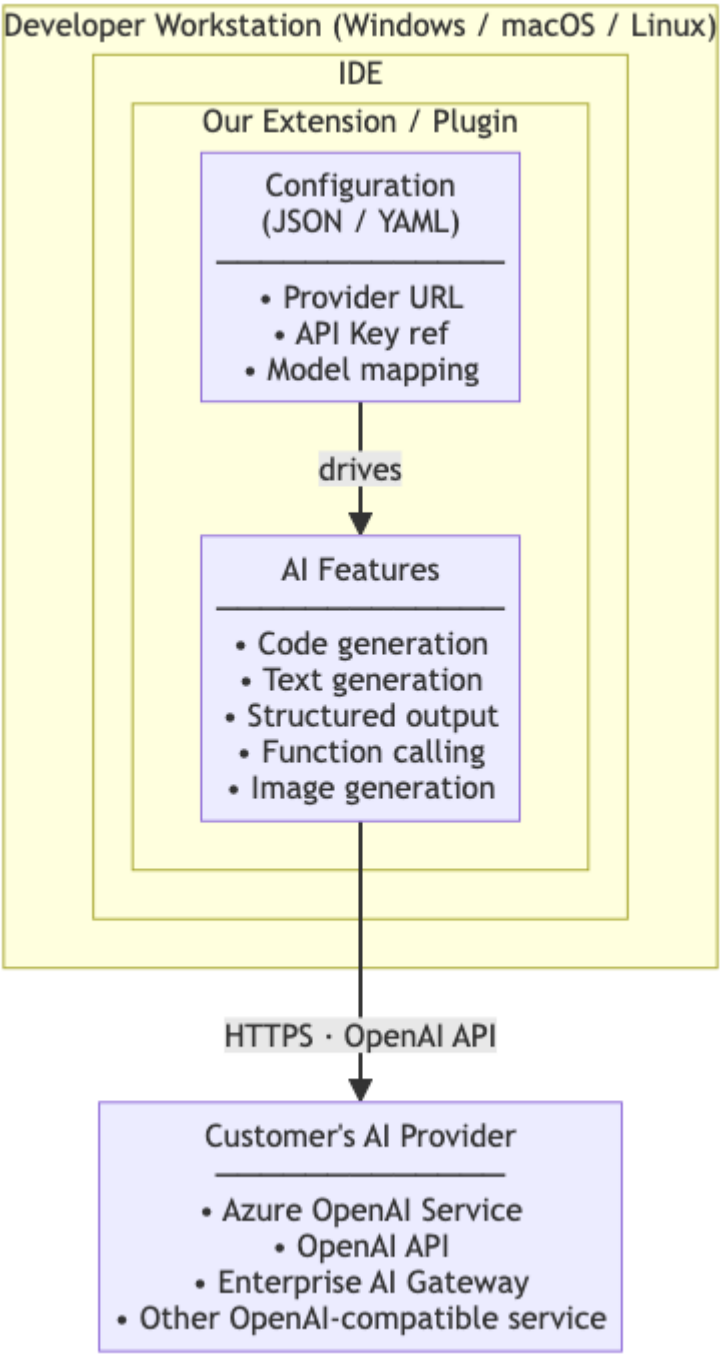
1. **An API key** (BYOK — Bring Your Own Key) for any OpenAI-compatible cloud service (OpenAI, Azure OpenAI, etc.)
2. **Network access** to that service from the developer's workstation

This "zero-infrastructure" evaluation path lets prospects go from installation to first AI interaction in minutes, not weeks—removing a common deal blocker in enterprise sales cycles where IT approval for infrastructure changes can stall evaluations indefinitely.

At the same time, the architecture provides a **clear graduation path to enterprise-grade AI governance**. When a customer moves from evaluation to production deployment, they can introduce an enterprise AI gateway, SSO, centralized policy

controls, and observability—all through configuration changes alone, with **zero code modifications** to the extension (see §7.4 — Gateway Graduation). This dual-track design ensures we win evaluations quickly while meeting the security and compliance bar required for enterprise procurement.

Deployment Model



Solution

This project establishes a standardized client-side configuration framework for accessing OpenAI API-compliant cloud services from our IDE extensions. Configuration follows IDE best practices (JSON/YAML) and supports:

- **Provider Flexibility:** Configure any OpenAI API-compatible endpoint (Azure OpenAI, OpenAI, Anthropic Claude, enterprise gateways, or self-hosted Ollama)
- **Enterprise Control:** IT administrators can deploy managed configuration profiles
- **User Customization:** Individual developers can override settings where permitted
- **Cross-Platform:** Consistent configuration across Windows, macOS, and Linux

Supported Use Cases

Use Case	API Endpoint / Protocol	Status
Text Generation	`/v1/chat/completions`	✅ In Scope
Code Generation	`/v1/chat/completions`	✅ In Scope
Structured Output	`/v1/chat/completions` (JSON mode)	✅ In Scope
Function Calling	`/v1/chat/completions` (tools)	✅ In Scope
Image Generation	`/v1/images/generations`	✅ In Scope (limited: single image generation only; batch generation, image editing, and variations not supported in v1)
Model Context Protocol (MCP)	MCP Server via `vscode.lm` API	✅ In Scope (v0.7)
BYOK / SecretStorage	VS Code `SecretStorage` API	✅ In Scope (v0.7)
Enterprise SSO	`vscode.authentication` provider	✅ In Scope (v0.7)
Agents / Assistants API	`/v1/assistants/*`	❌ Out of Scope
Embeddings	`/v1/embeddings`	⚠️ Future
Audio/Speech	`/v1/audio/*`	❌ Out of Scope

Value Proposition

Stakeholder	Value Delivered
Enterprise IT	Centrally manage AI provider configuration; enforce approved endpoints
Developers	AI features work out-of-the-box with their organization's approved provider
Security Teams	No hardcoded credentials; supports enterprise credential management
Our Product	Single codebase works with any OpenAI-compatible provider
Our Engineering	No per-provider code paths; configuration-driven provider selection

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2. Problem Statement & User Personas

Problem Statement

Our enterprise customers evaluating AI-powered features in our IDE extensions face configuration and control challenges:

Challenge	What Customers Tell Us	Our Impact
Provider Lock-in	"We have an Azure OpenAI contract; your tool must work with Azure, not just OpenAI"	Lost deals if we only support one provider
Configuration Complexity	"Every developer configures AI differently; we need consistency across 5,000 seats"	Support burden; inconsistent user experience
Credential Management	"We can't have API keys in plaintext settings files checked into repos"	Security review failures
No Central Control	"We need to push approved AI configurations to all developer machines"	IT cannot enforce policies
Cross-Platform	"Our developers use Windows, Mac, and Linux—config must work on all"	Platform-specific solutions are costly

The Opportunity

By building flexible, enterprise-manageable AI configuration:

- **Unlock Azure-first customers:** Large enterprises with existing Azure OpenAI deployments
- **Reduce support burden:** Consistent configuration patterns reduce "it doesn't work" tickets

- **Pass security reviews:** No hardcoded credentials; enterprise credential patterns supported
- **Enable IT control:** Managed configuration profiles for enterprise-wide deployment

User Personas

Customer-Side Personas

Persona	Role	Context	Primary Need
IT Administrator	Manages developer tooling	Rolling out our extension to 500+ developers	"Let me define one configuration and push it to all machines"
Security Engineer	Reviews developer tools	Evaluating our extension for approval	"Show me API keys are not stored in plaintext; show me the data flow"
Developer (Power User)	Uses AI features extensively	Wants to customize model settings	"Let me use a different model for certain tasks, if IT allows"
Developer (Typical)	Uses AI features casually	Just wants it to work	"I don't want to configure anything; it should just work with our company's AI"

Internal Personas

Persona	Role	Primary Need
Product Manager	Owns AI feature adoption	Unblock enterprise deals with flexible provider support
Extension Developer	Builds VS Code/Eclipse plugins	Clean API to call AI services without provider-specific code
Solutions Engineer	Supports enterprise customers	Clear configuration docs; managed deployment options
Technical Writer	Creates user/ admin docs	Configuration reference for all platforms and IDEs

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3. Goals & Success Metrics

Goals

1. **Provider Flexibility:** Support any OpenAI API-compatible service (Azure OpenAI, OpenAI, enterprise gateways) via configuration

2. **IDE-Native Configuration:** Configuration follows IDE best practices (VS Code `settings.json`, Eclipse preferences)
3. **Enterprise Manageability:** IT can deploy and manage configuration profiles across developer machines
4. **User Customization:** Developers can override settings where IT policy permits
5. **Cross-Platform Consistency:** Same configuration schema works on Windows, macOS, and Linux

Success Metrics

Metric	Baseline	Target	Measurement Method
Enterprise deals unblocked	0	15+ in Q1	CRM win/loss tracking (blocked by provider restrictions)
Time to first AI feature use	Varies	< 5 minutes	User telemetry (opt-in)
Configuration-related support tickets	N/A	< 5% of installs	Support ticket analysis
Security review pass rate	40%	95%	Enterprise security questionnaire approvals
Providers validated	1 (OpenAI)	4+	QA test matrix (Azure, OpenAI, vLLM, enterprise gateway)

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4. Assumptions & Dependencies

Assumptions

ID	Assumption	Risk if Invalid
A-01	Customers have access to an OpenAI API-compatible service (Azure OpenAI, OpenAI, or enterprise gateway)	Must provide guidance on provider setup
A-02	Developer workstations can reach the AI provider endpoint (directly or via corporate proxy)	Must support proxy configuration
A-03	Target providers implement OpenAI API v1 specification	Azure OpenAI ✅, OpenAI ✅, most enterprise gateways ✅
A-04	IT administrators have mechanisms to deploy configuration to developer	Central config management is optional enhancement

ID	Assumption	Risk if Invalid
	machines (GPO, MDM, config management)	
A-05	VS Code and Eclipse support our extension's settings schema	Standard extension settings mechanisms exist

Dependencies (Internal Engineering)

Dependency	Owner	Status	Risk if Delayed
Configuration schema design	Extension Team	🟡 In Progress	Cannot finalize settings documentation
VS Code settings integration	VS Code Team	🟢 Complete	Basic extension settings work
Eclipse preferences integration	Eclipse Team	🟡 In Progress	Eclipse users cannot configure AI
Credential storage via SecretStorage	Platform Team	🟢 Complete (v0.7)	Credentials stored in OS-level secure storage via VS Code API
HTTP client with proxy support	Platform Team	🟢 Complete	Corporate proxy environments blocked
Central config profile system	Platform Team	🔴 Not Started (v1.1)	No managed configuration option

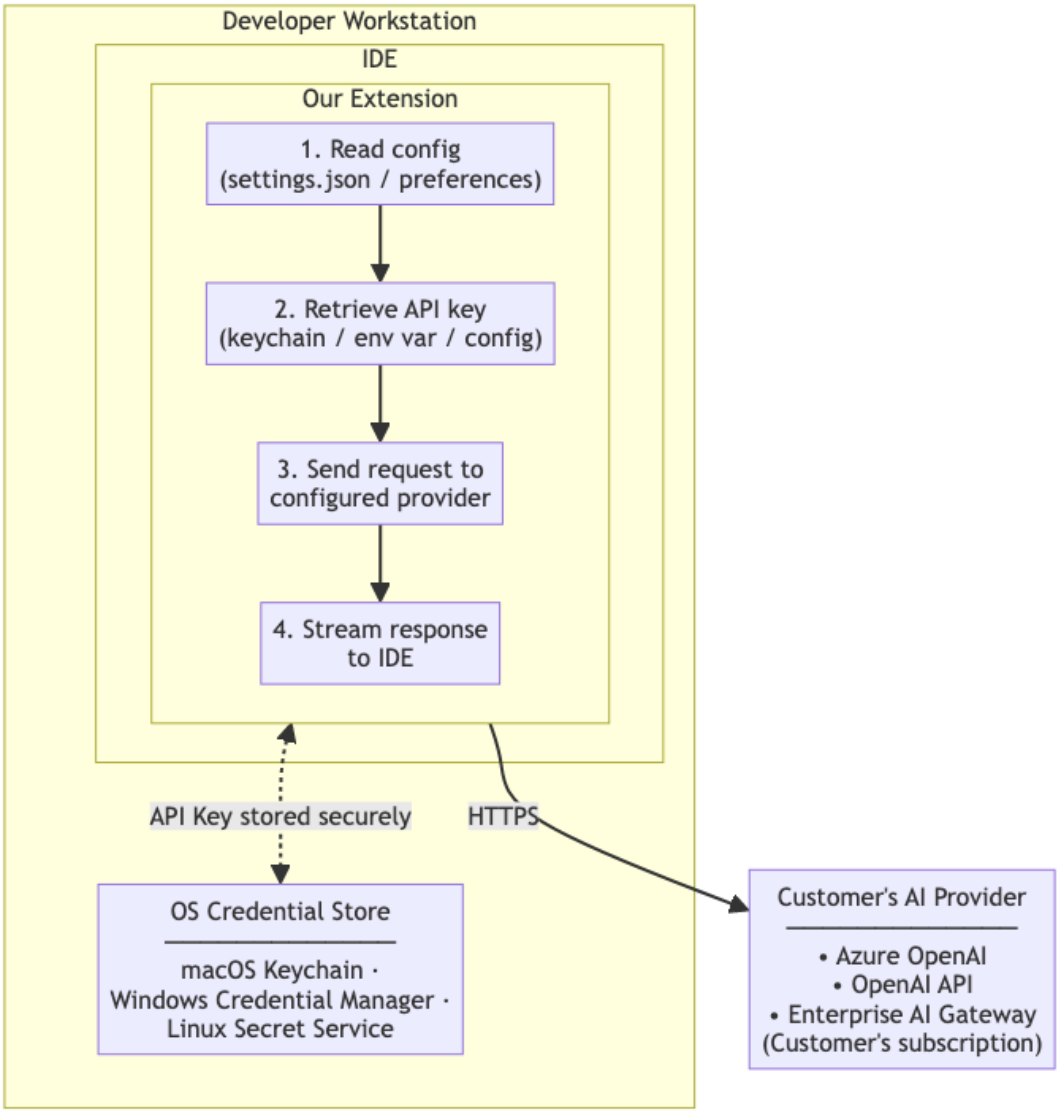
Customer-Side Prerequisites

Prerequisite	Customer Responsibility	We Provide
AI Provider Access	Provision Azure OpenAI or obtain OpenAI API access	Configuration examples per provider
API Credentials	Obtain API key from their provider	Secure storage guidance
Network Access	Ensure workstations can reach provider endpoint	Proxy configuration documentation
(Optional) Config Distribution	Deploy settings to developer machines	Settings schema documentation

5. Data Flow & Security

Data Flow Architecture

Our extension runs on the developer's workstation and communicates directly with the customer's configured AI provider:



What We (the Vendor) See

Data Type	Do We See It?	Notes
Prompts / Code	❌ No	Sent directly from workstation to customer's provider
Completions	❌ No	Returned directly from

Data Type	Do We See It?	Notes
		provider to workstation
API Keys	❌ No	Stored on customer's machine; never transmitted to us
Configuration	❌ No	Local to customer's machines
Usage Telemetry	⚠️ Only if opted-in	Optional, anonymized telemetry for product improvement

Security Considerations

Concern	How We Address
API Key Storage	Support OS credential stores (Keychain, Credential Manager, Secret Service); discourage plaintext in settings
Credential Leakage	API keys never logged; not included in error reports
Network Security	HTTPS required for provider communication; proxy support for corporate networks
Configuration Security	Settings can be scoped to user or workspace; sensitive fields marked appropriately

Enterprise Security Patterns

Pattern	Support
Corporate Proxy	HTTP_PROXY, HTTPS_PROXY environment variables; IDE proxy settings
Certificate Pinning	Custom CA certificate configuration for enterprise PKI
Credential Rotation	Credentials fetched fresh; no caching that would delay rotation
Audit Logging	Provider-side logging (Azure OpenAI logs, OpenAI usage dashboard)

6. Out of Scope (v1)

The following items are explicitly **not included** in the v1 release:

Explicitly Out of Scope

Item	Rationale	Future Version
Agents / Assistants API	Complex stateful interactions requiring persistent threads, file storage, and code interpreter runtime; fundamentally different architecture from v1 stateless tool-call model	Deferred indefinitely (requires separate architecture study)
LangGraph / Agent Frameworks	Local or remote agent runtimes (LangGraph, CrewAI, AutoGen) require stateful orchestration, Python runtimes, and human-in-the-loop UX patterns beyond v1 scope. The v1 tool-call loop (model-driven, stateless) provides agent-like behavior without an agent runtime. Architectural hedges in §7.6 preserve future optionality.	v2+
Audio/ Speech APIs	`/v1/audio/*` endpoints not needed for dev tools	N/A
Embeddings API	Search/RAG use cases deferred	v1.1

Deferred to Future Versions

Item	Rationale	Future Version
Central Configuration Management	Enterprise profile distribution from control point	v1.1
Admin Console / UI	Configuration via JSON/ YAML sufficient for v1	v1.1
AWS Bedrock Native Auth (SigV4)	Bedrock's native authentication uses AWS SigV4 signing, which requires an AWS SDK dependency. Customers	v1.1

Item	Rationale	Future Version
	can access Bedrock models via an enterprise AI gateway or Bedrock's OpenAI-compatible endpoint in v1.	
Usage Analytics Dashboard	Customers use provider's own dashboards	v2
Prompt Templates / Library	Would require content storage	v2

Customer Responsibility

Item	Notes
Content Filtering	Configured at provider level (Azure Content Safety, OpenAI moderation)
Usage Monitoring	Use provider dashboards (Azure Portal, OpenAI Usage)
Rate Limiting	Provider-side limits; customer manages quotas with their provider

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7. VS Code Extension Design-Phase Requirements

> **Supersedes:** This section refines and supersedes earlier guidance on provider integration, credential storage, and tool/agent capabilities from prior SRS versions. Where conflicts exist, the requirements below take precedence.
>
> **Specifically supersedes:** - REQ-07 (OS credential stores) — now clarified as fallback to SecretStorage (REQ-52) - REQ-10 (no plaintext keys) — now refined as REQ-53 (no globalState/workspaceState) - REQ-14 (pass tools unmodified) — now refined as REQ-37/REQ-38 (tool definition and tool_choice support) - REQ-17 (proxy support) — now promoted to Must as REQ-63 with explicit HTTP_PROXY/HTTPS_PROXY env var support

As the development team moves into the **design phase** for the VS Code extension, the following requirements ensure the product remains **service-agnostic**, **secure**, and **ready for enterprise "graduation"** from evaluation to production.

7.1 Provider-Agnostic Interface (Wire Protocol)

Requirement: The extension must use a **Universal LLM Client** that communicates via the OpenAI Chat Completions wire format.

Implementation:

- **No vendor-specific SDKs** as the primary integration path. Do not depend on `@openai/openai` or `@anthropic-ai/sdk` directly. Use plain HTTP (Node.js `fetch` or equivalent) against the OpenAI-compatible REST surface.
- Target a configurable **`BASE_URL`** and **`MODEL_NAME`**.
- By default, `BASE_URL` should be editable in VS Code Settings (e.g., `https://api.openai.com/v1` for evaluation or `https://gateway.acme.com/v1` for enterprise).
- **URL Path Handling:** The `BASE_URL` must be the **complete base path** to which API-specific paths (e.g., `/chat/completions`, `/images/generations`) are appended. For Azure OpenAI, this includes the deployment: `https://<resource>.openai.azure.com/openai/deployments/<deployment-name>`. The extension appends `/chat/completions` (or other endpoint paths) to construct the full request URL.
- The extension must pass through custom headers and query parameters (e.g., Azure `api-version`) without hard-coding provider-specific logic.

7.2 Frictionless BYOK — Secure Key Storage

Requirement: Enable users to supply their own API keys without requiring backend infrastructure.

Implementation:

- Utilize the **VS Code `SecretStorage` API** (`context.secrets`) to persist API keys locally on the user's machine. **Never** store keys in `globalState`, `workspaceState`, or standard JSON settings files.
- Support **Header Injection**: The extension must allow custom authentication headers (e.g., `X-Api-Key`, `Authorization: Bearer …`) to be dynamically populated from `SecretStorage` at request time.
- Provide a first-run "Set API Key" command that writes to `SecretStorage` and confirms success.

7.3 Tooling & Agency (Model Context Protocol)

Requirement: Use the **Model Context Protocol (MCP)** as the standard for tool capabilities. **No local AI agent runtime is required.**

Dual MCP Role:

The extension serves two complementary MCP roles:

1. **MCP Server** — The extension registers as an MCP Server (via `vscode.lm.registerMcpServerDefinitionProvider`), exposing tools that *other* MCP hosts in the VS Code ecosystem (e.g., GitHub Copilot) can discover and invoke.
2. **Tool-Call Orchestrator** — For its *own* LLM interactions, the extension contains a lightweight orchestration loop that iteratively executes `tool_calls` returned by the cloud model via the Chat Completions API until the model produces a final text response.

Critically, the cloud-hosted model (e.g., GPT-4, Claude) acts as its own planner. The extension is a **stateless executor** — it does not run a local agent, reasoning framework, or planning engine. "Agent-like" multi-step behavior emerges from the model's native ability to emit `tool_calls` iteratively, with the extension mechanically executing and returning results.

Implementation:

- Register the extension as an **MCP Server** using the `vscode.lm.registerMcpServerDefinitionProvider` API.
- Expose extension-specific capabilities as **MCP Tools**, including (but not limited to):
  - `search_codebase` — workspace-wide text / symbol search
  - `run_unit_tests` — execute test suites and return results
  - `read_file_context` — read file contents with line ranges
- **MCP Tool Permission Model:** Tools exposed via MCP Server must be classified by risk level (`safe`, `read-only`, `write-only`, `destructive`). The extension must maintain an allowlist of external MCP hosts authorized to invoke each tool category. By default, only `safe` and `read-only` tools are externally visible. Destructive tools (e.g., `run_terminal_command`) are internal-only unless explicitly authorized.
- **Workspace Trust Integration:** In untrusted workspaces (per VS Code Workspace Trust API), the extension must disable tool execution, limit model interactions to read-only context (e.g., open file contents only), and suppress `run_unit_tests` and `run_terminal_command` tools entirely.
- **Reasoning vs. Execution model:** The cloud model is the brain (decides *what* to do). The extension is the hands (decides *how* to do it safely). The tool-call orchestration loop sends tool definitions to the model, executes any returned `tool_calls` locally via the VS Code Extension API, appends results as `tool` role messages, and re-calls the model — repeating until a final text response is produced or a maximum iteration limit is reached.
- **Tool Execution Path:** When the extension's own tool-call orchestrator invokes tools for its internal LLM interactions, it calls tool implementations **directly** (internal function calls), not via the MCP protocol. The MCP Server registration is solely for exposing tools to *external* MCP hosts (e.g., GitHub Copilot). This avoids protocol overhead and serialization round-trips for internal tool use.
- MCP tool definitions must include JSON Schema parameter descriptions to maximize model accuracy.
- The tool-call orchestration loop must be implemented as a **modular, replaceable component** (`ToolOrchestrator` interface) to support future substitution with more advanced orchestration backends (see §7.6).

7.4 Path to Enterprise — The Gateway Graduation

Requirement: The architecture must support a **zero-code-change transition** from direct SaaS access to enterprise-governed access.

Implementation:

- **SSO Integration:** Implement support for the `vscode.authentication` provider API. When a `BASE_URL` points to an enterprise gateway, the extension should attempt to exchange a VS Code session token for a gateway bearer token via OIDC / OAuth flow.

- **Telemetry & Observability:** Include a `X-Request-Id` header (UUID v4) in **all** outgoing LLM calls to allow enterprise IT to trace requests from the IDE through their AI Gateway and into the upstream provider.
- **Proxy awareness:** Respect VS Code's built-in proxy settings and `HTTP_PROXY` / `HTTPS_PROXY` environment variables for corporate network environments.

7.5 Evaluation ⊡ Enterprise Specification Summary

Component	Evaluation Requirement (BYOK)	Enterprise Requirement (Production)
Auth	User-pasted key in `SecretStorage`	OIDC / OAuth via `vscode.authentication`
Routing	Direct to SaaS (OpenAI, Anthropic, Azure OpenAI)	Internal AI Gateway (Kong, Cloudflare, etc.)
Protocol	OpenAI-Compatible REST + MCP for Tools	OpenAI-Compatible REST + MCP for Tools
Connectivity	Public Internet (Node.js `fetch`)	Corporate VPN / Proxy aware
Telemetry	Optional opt-in	`X-Request-Id` tracing through gateway
Key Storage	VS Code `SecretStorage`	`SecretStorage` + SSO token exchange

7.6 Future-Proofing: Architectural Hedges for Agent Frameworks

> **Context:** While local/remote agent frameworks (e.g., LangGraph, CrewAI) are explicitly **out of scope** for v1 (see §6), the following low-cost architectural hedges ensure the extension can adopt advanced orchestration in the future with minimal refactoring. These are *design constraints*, not features — they impose structure on v1 implementation to preserve optionality.

1. Modular Tool Orchestrator

- The tool-call loop (send tools → receive `tool_calls` → execute → return results → repeat) must be encapsulated behind a **`ToolOrchestrator` interface**. The v1 implementation is a simple iterative loop driven by Chat Completions `tool_calls`. In a future version, this interface could be swapped for a LangGraph-backed orchestrator or cloud-hosted agent endpoint — with no changes to the rest of the extension.

2. Conversation Thread IDs

- Every conversation or multi-turn interaction must be assigned a **`thread_id`** (UUID v4), even though v1 does not persist conversation state. This in-memory correlation ID maps directly to LangGraph's thread concept and enables future checkpointing, resume, and replay without retroactively adding identifiers.

3. Event-Typed Streaming

- The internal streaming layer must emit **typed events** rather than raw token strings. The v1 event types are: `token`, `tool_call`, `tool_result`, `status`, and `error`. Even though v1 consumers primarily use `token` events, this structure allows future agent orchestrators to emit additional event types (e.g., `plan_step`, `human_approval_needed`, `state_transition`) without breaking the streaming contract.

4. Multi-Step UX Readiness

- The UI layer must not assume that a single user action produces a single model response. The tool-call loop already breaks this assumption, but the UX components (progress indicators, output panels) must be designed to display **step-by-step progress** for multi-turn interactions. This ensures future agent workflows — which may involve planning, branching, and human-in-the-loop approvals — can surface their state in the existing UI framework.

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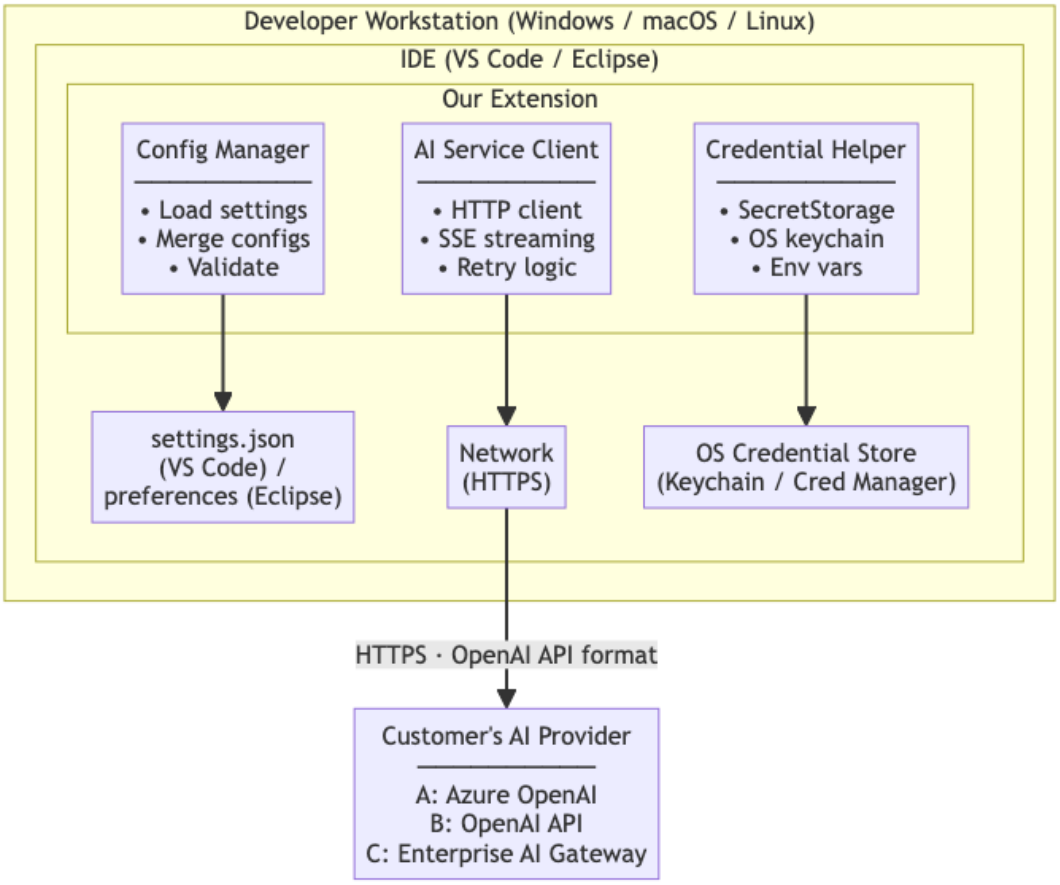
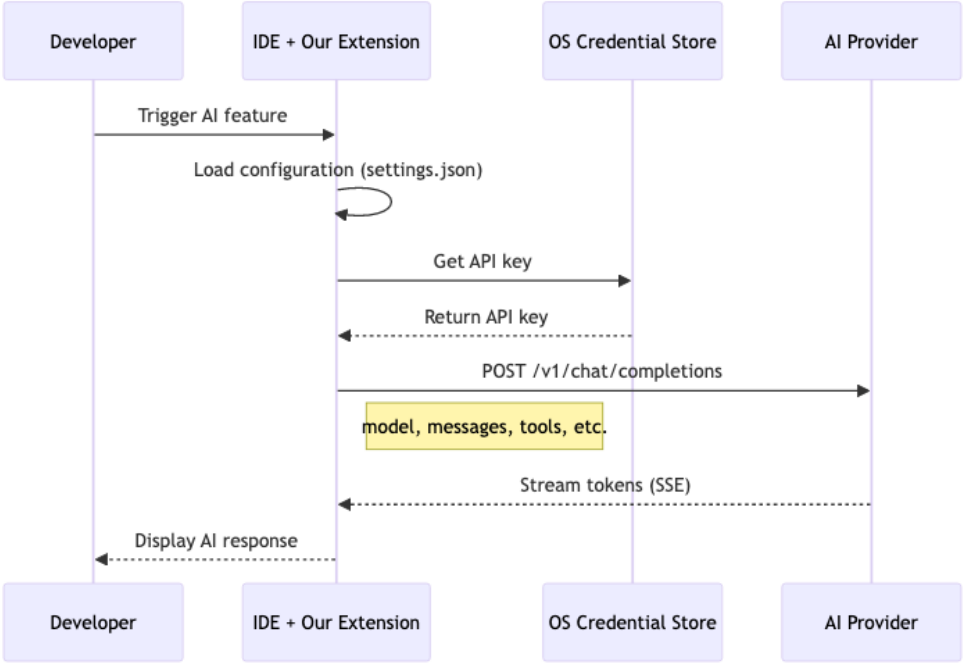
8. System Architecture

Component Overview

Our product is an **IDE extension** installed on developer workstations. It communicates directly with the customer's configured AI provider via the **Universal LLM Client** (see §7.1)—no intermediate backend or gateway required (though customers may optionally use an enterprise AI gateway).



Request Flow



Architecture Decisions

Decision	Choice	Rationale
Runtime	IDE extension (no separate backend)	Simple deployment; standard extension install
LLM integration	Universal LLM Client via OpenAI-compatible REST (no vendor SDKs)	Service-agnostic; zero-code-change provider swap (§7.1)
Configuration	IDE-native settings (JSON/YAML)	Familiar to developers; supports IDE config sync
Credential storage	VS Code `SecretStorage` API (preferred); OS credential store fallback	Security best practice; avoids plaintext keys (§7.2)
Auth header	Dynamic header injection from `SecretStorage`	Supports `Authorization`, `X-Api-Key`, and custom headers
Provider communication	Direct HTTPS to provider (or enterprise gateway)	No middleman; simplest data flow; gateway-ready (§7.4)
Streaming	SSE (Server-Sent Events)	Real-time token display; standard OpenAI pattern
Agentic capabilities	Model Context Protocol (MCP) via `vscode.lm` API	Industry-standard tool protocol; hands/brain separation (§7.3)
Enterprise auth	`vscode.authentication` provider for SSO/OIDC	Zero-code-change graduation to enterprise gateway (§7.4)
Observability	`X-Request-Id` header on all outgoing calls	End-to-end tracing through enterprise AI gateways
Cross-platform	Single codebase for Win/Mac/Linux	Extension APIs abstract platform differences

Provider Type Behaviors

The `type` field in the provider configuration controls provider-specific behaviors:

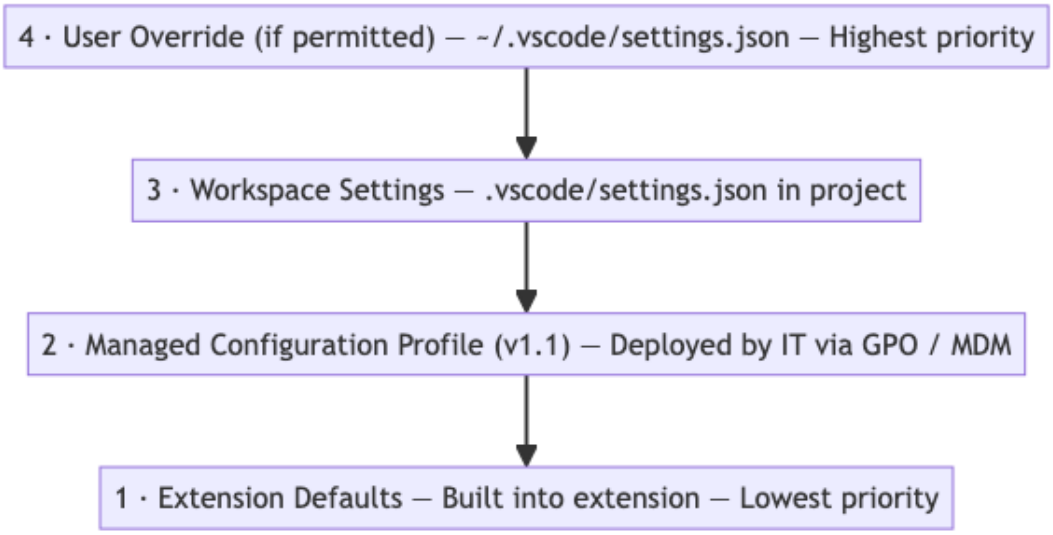
Provider Type	Auth Header	Path Construction	Default Headers	Notes
`azure-openai`	`api-key`	`baseUrl` + `/chat/completions` + `?api-version=X`	`api-key: <from SecretStorage>`	Requires `api-version` in headers or query params
`openai`	`Authorization`	`baseUrl` + `/chat/completions`	`Authorization: Bearer <from SecretStorage>`	Standard OpenAI format

Provider Type	Auth Header	Path Construction	Default Headers	Notes
`anthropic`	`x-api-key`	`baseUrl` + `/chat/completions`	`x-api-key: <from SecretStorage>` `anthropic-version: 2023-06-01`	Uses Anthropic's OpenAI-compatible endpoint
`ollama`	None (optional)	`baseUrl` + `/chat/completions`	None	No auth by default; supports optional Bearer token if Ollama is secured
`openshift-ai`	`Authorization`	`baseUrl` + `/chat/completions`	`Authorization: Bearer <from SecretStorage>`	Uses Kubernetes service account or OAuth token
`bedrock`	N/A (v1.1)	N/A	N/A	Native SigV4 auth deferred; use via gateway or OpenAI-compatible endpoint
`compatible`	Configurable via `authHeaderName`	`baseUrl` + `/chat/completions`	User-specified via `authHeaderName` and `authMode`	Generic fallback for any OpenAI-compatible provider

Configurability: Users can override the default auth header using the `authHeaderName` field in the provider configuration. When `authMode: "sso"` is set, the `vscode.authentication` provider is used instead of `SecretStorage`.

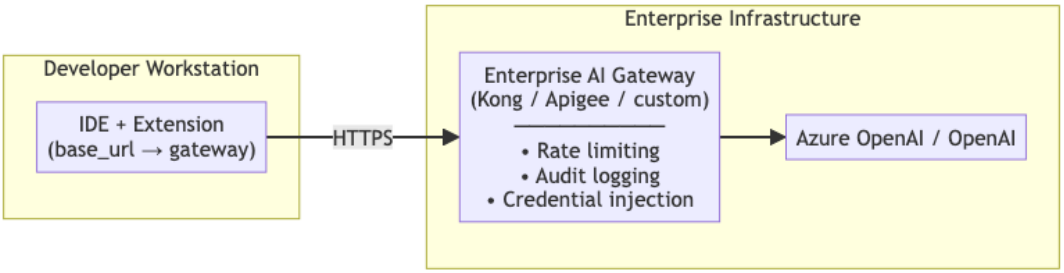
Configuration Hierarchy

Configuration can be set at multiple levels, with higher levels overriding lower:



Optional: Enterprise AI Gateway

Some enterprises deploy a central AI gateway. Our extension supports this pattern:



Supported Providers (v1)

Provider	Compatibility	Configuration Notes
Azure OpenAI	✅ Native	Set `base_url` to deployment; add `api-version` header
OpenAI	✅ Native	Default; just provide API key
Enterprise Gateways	✅ Native	Point `base_url` to gateway; gateway handles upstream
Ollama	✅ Native	Self-hosted; point `base_url` to local instance (e.g., `http://localhost:11434/v1`)
Anthropic Claude	✅ Via OpenAI-compatible endpoint	Set `base_url` to `https://api.anthropic.com/v1/`; BYOK with Anthropic API key. Uses Anthropic's OpenAI-compatible surface — no adapter or translation layer needed.
Red Hat OpenShift AI	✅ Via OpenAI-compatible endpoint	Set `base_url` to the OpenShift AI model serving route (e.g., `https://<model-`

Provider	Compatibility	Configuration Notes
		route>.apps.<cluster>/v1). Uses KServe / vLLM runtime which exposes an OpenAI-compatible inference API. BYOK with the serving endpoint's bearer token.
AWS Bedrock	⚠️ Conditional	Supported when accessed through Bedrock's OpenAI-compatible endpoint or an enterprise AI gateway that exposes an OpenAI-compatible surface. Native Bedrock SigV4 auth is **not** supported in v1; customers requiring direct Bedrock access without a gateway should use the gateway graduation path (§7.4).

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9. Prompt Engineering & API Patterns

This section defines how developers building features within our extensions should construct and manage AI interactions. These patterns ensure consistency, reusability, and reliable outputs.

9.1 Message Roles

The OpenAI Chat Completions API uses distinct message roles. Our extension SDK must properly support all roles:

Role	Purpose	When to Use
system	Sets behavior, persona, and constraints for the model	Once at the start of each conversation/request; defines "who" the AI is
user	Contains the actual user request or input	For prompts derived from user actions or code context
assistant	Contains previous AI responses	For multi-turn conversations; providing examples (few-shot)

Example: Programmatic Prompt Construction

```
// Code review feature - programmatic prompt (not interactive
        chat)
const messages: ChatMessage[] = [
  {
    role: "system",
    content: `You are a senior software engineer performing code
        review.
Focus on: security vulnerabilities, performance issues, and
        maintainability.
```

```
Be concise. Use bullet points. Do not rewrite the code unless
        asked.`
  },
  {
    role: "user",
    content: `Review this code:\n\n\`\`\`${language}\n$
        {codeSnippet}\n\`\`\``
  }
];
```

9.2 Prompt Templates

For reusable, programmatic prompts (not user-authored chat), the extension should support templated prompts:

Template Structure

```yaml
# Prompt template: code-review.yaml
id: code-review
name: "Code Review"
description: "Analyze code for issues and improvements"
version: "1.0"

system_prompt: |
  You are a senior software engineer performing code review.
  Focus on: {{focus_areas}}
  Language: {{language}}
  Be concise. Use bullet points.

user_prompt_template: |
  Review this {{language}} code:

  ```{{language}}
  {{code}}
```

{{#if specific_concern}} Pay special attention to: {{specific_concern}} {{/if}}

parameters: - name: language type: string required: true - name: code type: string required: true - name: focus_areas type: string default: “security, performance, maintainability” - name: specific_concern type: string required: false

model_settings: temperature: 0.3 # Lower for more consistent analysis max_tokens: 1024

```
**Programmatic Usage**

```typescript
const result = await ai.executeTemplate("code-review", {
  language: "typescript",
  code: selectedText,
  focus_areas: "security, error handling",
```

```
  specific_concern: "SQL injection"
});
```

9.3 Instruction Following Best Practices

To ensure reliable model outputs, prompts should follow these patterns:

Pattern	Description	Example
Role Definition	Clearly define who/what the model is	"You are a TypeScript expert…"
Task Specification	Explicitly state the task	"Your task is to refactor this function to…"
Output Format	Specify desired format	"Respond with a JSON object containing…"
Constraints	Define what NOT to do	"Do not include explanations. Only output code."
Examples (Few-shot)	Provide input/output examples	Include 1-3 examples in system or user message

Anti-patterns to Avoid

Anti-pattern	Problem	Better Approach
Vague instructions	Inconsistent outputs	Be specific: "List exactly 3 issues"
No output format	Hard to parse results	Always specify format
Overly long prompts	Token waste, confusion	Keep focused; use separate calls for different tasks
Prompt injection risk	User input may override instructions	Clearly delimit user content with markers

9.4 Structured Output Handling

For features requiring parseable responses, use the `response_format` parameter or explicit JSON instructions:

Method 1: JSON Mode (Recommended)

```
const response = await ai.complete({
  messages: [...],
  response_format: { type: "json_object" },
  // When using JSON mode, instruct the model what JSON to produce
});
```

```
// System prompt must mention JSON:
// "Respond with a JSON object containing: { issues: [...],
        summary: string }"
```

Method 2: Explicit Schema in Prompt

```
const systemPrompt = `
Analyze the code and respond with this exact JSON structure:
{
  "issues": [
    {
      "severity": "high" | "medium" | "low",
      "line": number,
      "description": string,
      "suggestion": string
    }
  ],
  "overall_quality": "good" | "needs_work" | "poor",
  "summary": string
}

Respond ONLY with valid JSON. No additional text.
`;
```

Response Parsing

```
interface CodeReviewResult {
  issues: Array<{
    severity: "high" | "medium" | "low";
    line: number;
    description: string;
    suggestion: string;
  }>;
  overall_quality: "good" | "needs_work" | "poor";
  summary: string;
}

const result = await ai.complete<CodeReviewResult>({
  messages,
  response_format: { type: "json_object" },
  responseSchema: CodeReviewResultSchema  // Runtime validation
});
```

9.5 Function Calling (Tools)

For features that need the model to invoke defined functions:

Defining Tools

```
const tools: Tool[] = [
  {
    type: "function",
    function: {
```

```
      name: "get_file_contents",
      description: "Read the contents of a file in the workspace",
      parameters: {
        type: "object",
        properties: {
          file_path: {
            type: "string",
            description: "Relative path to the file"
          }
        },
        required: ["file_path"]
      }
    }
  },
  {
    type: "function",
    function: {
      name: "run_terminal_command",
      description: "Execute a terminal command. **WARNING:** This
        tool executes arbitrary commands and is classified as
        DESTRUCTIVE. Requires explicit user approval when
        `requireApprovalForDestructiveTools` is enabled. Not
        exposed to external MCP hosts by default.",
      parameters: {
        type: "object",
        properties: {
          command: { type: "string", description: "Shell command
        to execute" },
          working_directory: { type: "string", description:
        "Working directory (optional)" }
        },
        required: ["command"]
      }
    }
  }
];
```

Handling Tool Calls

```
const response = await ai.complete({
  messages,
  tools,
  tool_choice: "auto"  // or "required" or { type: "function",
        function: { name: "..." } }
});

if (response.tool_calls) {
  for (const toolCall of response.tool_calls) {
    const args = JSON.parse(toolCall.function.arguments);

    // Execute the tool
    const result = await executeFunction(toolCall.function.name,
        args);
```

```
    // Send result back to model
    messages.push(
      { role: "assistant", content: null, tool_calls:
        [toolCall] },
      { role: "tool", tool_call_id: toolCall.id, content:
        JSON.stringify(result) }
    );
  }

  // Continue conversation with tool results
  const finalResponse = await ai.complete({ messages, tools });
}
```

9.6 Context Management

For features that incorporate workspace context (code, files, symbols):

Context Type	How to Include	Token Considerations
Selected Code	Include in user message with language markers	Count tokens; truncate if needed
File Contents	Include relevant portions, not entire files	Summarize or chunk large files
Symbol Definitions	Include type signatures, interfaces	Prioritize relevant symbols
Error Messages	Include full error with stack trace	May truncate very long traces
Documentation	Include relevant doc snippets	Use RAG pattern for large docs

Context Prioritization

```
interface ContextBuilder {
  // Add context with priority (higher = more important to keep)
  addContext(content: string, priority: number, label: string):
        void;

  // Build final context respecting token budget
  build(maxTokens: number): string;
}

const context = new ContextBuilder();
context.addContext(errorMessage, 100, "error");      // Highest
        priority
context.addContext(selectedCode, 90, "selection");    // High
context.addContext(fileContents, 50, "file");         // Medium
context.addContext(relatedTypes, 30, "types");        // Lower

const finalContext = context.build(4000);  // Fits in context
        window
```

10. Functional Requirements (EARS Syntax)

Configuration

ID	Priority	Requirement (EARS Format)
REQ-01	Must	**The extension shall** read configuration from IDE-native settings (VS Code `settings.json`, Eclipse preferences).
REQ-02	Must	**The extension shall** support configuration in both JSON and YAML formats where the IDE permits.
REQ-03	Must	**The extension shall** allow configuration of `base_url` to any OpenAI API-compatible endpoint.
REQ-04	Must	**The extension shall** support model ID aliasing, mapping user-friendly names to provider model identifiers.
REQ-05	Must	**The extension shall** support configuration of custom headers (e.g., `api-version` for Azure OpenAI).
REQ-06	Should	**When** both user and workspace settings exist, **the extension shall** merge them with **workspace settings taking precedence** per VS Code convention (user settings as fallback).

Credential Management

ID	Priority	Requirement (EARS Format)
REQ-07	Must	**The extension shall** use VS Code `SecretStorage` API as the primary credential storage mechanism; **when** SecretStorage is unavailable (e.g., headless/container environments), **the extension shall** fall back to OS credential stores (macOS Keychain, Windows Credential Manager, Linux Secret Service) with a warning to the user about reduced security.
REQ-07a	Must	**When** both SecretStorage and OS credential stores are unavailable, **the extension shall** disable AI features requiring authentication and display a persistent warning: "Credential storage unavailable. AI features disabled. Use environment variables (OPENAI_API_KEY) as workaround."
REQ-08	Must	**The extension shall** support retrieving API keys from environment variables.
REQ-09	Should	**The extension shall** support API key entry via IDE-native secret input (masked input fields).

ID	Priority	Requirement (EARS Format)
REQ-10	Must	**The extension shall not** store API keys in plaintext in settings files by default.
REQ-11	Must	**The extension shall** never include API keys in logs, error reports, or telemetry.

API Integration

ID	Priority	Requirement (EARS Format)
REQ-12	Must	**The extension shall** implement the OpenAI Chat Completions API (`/v1/chat/completions`).
REQ-13	Must	**The extension shall** support streaming responses via Server-Sent Events (SSE).
REQ-14	Must	**When** a request includes `tools` or `functions` parameters, **the extension shall** pass them unmodified to the provider.
REQ-15	Must	**When** `response_format: { "type": "json_object" }` is specified, **the extension shall** include it in the request.
REQ-16	Should	**The extension shall** support the Images API (`/v1/images/generations`) for image generation features.
REQ-17	Must	**The extension shall** support HTTP/HTTPS proxy configuration via IDE proxy settings and/or environment variables.
REQ-93	Must	**The extension shall** enforce network timeouts: 10 seconds for connection establishment, 60 seconds for streaming reads, 30 seconds for non-streaming reads.
REQ-94	Must	**When** a provider response exceeds 1MB, **the extension shall** truncate the response and display a warning: “Response truncated at 1MB. Consider using streaming or reducing request scope.”

Concurrent Request Handling

ID	Priority	Requirement (EARS Format)
REQ-79	Must	**The extension shall** support multiple concurrent chat sessions without blocking or interference.
REQ-80	Should	**When** multiple requests are initiated simultaneously, **the extension shall** queue them and enforce per-provider rate limits to prevent throttling errors.

Provider Capability Detection

ID	Priority	Requirement (EARS Format)
REQ-81	Should	**The extension shall** detect provider support for vision/multimodal inputs via `/v1/models` endpoint or provider-specific capability metadata.
REQ-82	Should	**When** a provider does not support a requested feature (e.g., tool calling, vision), **the extension shall** gracefully degrade by disabling the feature and notifying the user.
REQ-83	Should	**The extension shall** cache provider capability information for the duration of the session to minimize redundant API calls.

User Experience

ID	Priority	Requirement (EARS Format)
REQ-18	Should	**When** configuration is incomplete, **the extension shall** prompt the user with guided setup.
REQ-19	Should	**The extension shall** provide a "Test Connection" command to validate provider connectivity.
REQ-20	Must	**When** an API error occurs, **the extension shall** display a user-friendly error message with troubleshooting guidance.
REQ-21	Must	**The extension shall** indicate streaming progress during AI response generation.
REQ-21a	Should	**When** a user is waiting for an AI response, **the extension shall** display a progress indicator showing elapsed time and estimated completion if streaming has not yet started.

Accessibility & Internationalization

ID	Priority	Requirement (EARS Format)
REQ-95	Must	**The extension shall** support keyboard-only operation: all AI commands accessible via Command Palette (VS Code) or menu shortcuts, no mouse-only features.
REQ-96	Must	**When** streaming AI responses, **the extension shall** announce status changes to screen readers ("Thinking…", "Generating response…", "Response complete") using ARIA live regions or platform equivalents.
REQ-97	Should	**The extension shall** use an i18n framework (e.g., `vscode-nls`, Eclipse NLS) with externalized strings in resource bundles, enabling future localization without code changes.

ID	Priority	Requirement (EARS Format)
REQ-98	Should	**The extension shall** ship with English (en-US) locale in v1; message keys shall follow convention `extension.category.message` for easy translation in v1.1+.

Multi-Provider Configuration

ID	Priority	Requirement (EARS Format)
REQ-84	Should	**The extension shall** support configuration of multiple named provider profiles (e.g., "dev-azure", "prod-openai") in user or workspace settings.
REQ-85	Should	**The extension shall** provide a UI command or quick pick to switch the active provider profile at runtime.

Enterprise Management (v1.1)

ID	Priority	Requirement (EARS Format)
REQ-22	Won't (v1)	**The extension shall** support loading configuration profiles from a central management endpoint.
REQ-23	Won't (v1)	**When** managed configuration is deployed, **the extension shall** allow IT to lock certain settings from user override.

> **Note:** REQ-24 through REQ-27 were removed during the v0.6 refactoring when prompt engineering requirements were reorganized. The requirement numbering gap is preserved for historical traceability.

Prompt Construction & Message Roles

ID	Priority	Requirement (EARS Format)
REQ-28	Must	**The extension SDK shall** support all OpenAI message roles: `system`, `user`, `assistant`, and `tool`.
REQ-29	Must	**The extension SDK shall** provide a programmatic API for constructing message arrays without requiring user interaction.
REQ-30	Should	**The extension SDK shall** support prompt templates with variable substitution for reusable, parameterized prompts.
REQ-31	Should	**When** constructing prompts, **the extension SDK shall** provide utilities to estimate token counts before sending requests.
REQ-32	Should	**The extension SDK shall** support context prioritization, allowing developers to specify

ID	Priority	Requirement (EARS Format)
		which context to retain when truncating for token limits.

Context Window Overflow Handling

ID	Priority	Requirement (EARS Format)
REQ-86	Should	**When** the constructed prompt exceeds 90% of the model's known context window, **the extension shall** warn the user before sending the request.
REQ-87	Should	**When** a provider returns a token limit error (e.g., HTTP 400 with `context_length_exceeded`), **the extension shall** automatically truncate the oldest non-system messages and retry once.
REQ-88	Should	**When** automatic truncation occurs, **the extension shall** notify the user that context was reduced and log which messages were removed.

Structured Output

ID	Priority	Requirement (EARS Format)
REQ-33	Must	**The extension SDK shall** support the `response_format: { type: "json_object" }` parameter for JSON mode.
REQ-34	Must	**When** JSON mode is requested, **the extension SDK shall** validate that the system prompt instructs the model to produce JSON.
REQ-35	Should	**The extension SDK shall** provide typed response parsing with runtime validation against a provided schema.
REQ-36	Should	**When** JSON parsing fails, **the extension SDK shall** provide a clear error indicating the parsing failure and include the raw response for debugging.

Function Calling (Tools)

ID	Priority	Requirement (EARS Format)
REQ-37	Must	**The extension SDK shall** support defining tools/functions with JSON Schema parameter definitions.
REQ-38	Must	**The extension SDK shall** support all `tool_choice` options: `auto`, `required`, `none`, and specific function selection.
REQ-39	Must	

ID	Priority	Requirement (EARS Format)
		When the model returns `tool_calls`, **the extension SDK shall** provide the parsed function name and arguments.
REQ-40	Must	**The extension SDK shall** support sending tool results back to the model via the `tool` role message.
REQ-41	Should	**The extension SDK shall** provide a tool execution loop helper that handles multi-turn tool calling until completion.

Security & Privacy

ID	Priority	Requirement (EARS Format)
REQ-42	Must	**The extension shall** use HTTPS for all provider communication.
REQ-43	Must	**The extension shall** support custom CA certificates for enterprise PKI environments.
REQ-44	Must	**The extension shall not** send telemetry without explicit user opt-in.
REQ-45	Must	**The extension shall not** persist prompts, completions, or conversation history to disk (unless explicitly enabled for a feature).
REQ-46	Must	**When** constructing prompts with user-provided content, **the extension SDK shall** clearly delimit user content to reduce prompt injection risk.

Provider-Agnostic Interface (§7.1)

ID	Priority	Requirement (EARS Format)
REQ-47	Must	**The extension shall** use a Universal LLM Client that communicates exclusively via the OpenAI Chat Completions wire format (`/v1/chat/completions`).
REQ-48	Must	**The extension shall not** depend on vendor-specific SDKs (e.g., `@openai/openai`, `@anthropic-ai/sdk`) as the primary integration path.
REQ-49	Must	**The extension shall** expose a configurable `BASE_URL` setting that defaults to `https://api.openai.com/v1` and is editable in VS Code Settings.
REQ-50	Must	**The extension shall** expose a configurable `MODEL_NAME` setting to specify the target model identifier.
REQ-51	Must	

ID	Priority	Requirement (EARS Format)
		The extension shall pass through custom headers and query parameters to the provider without hard-coding provider-specific logic.

Secure Key Storage — BYOK (§7.2)

ID	Priority	Requirement (EARS Format)
REQ-52	Must	**The extension shall** use the VS Code `SecretStorage` API (`context.secrets`) to persist API keys locally on the user's machine.
REQ-52a	Must	**The extension shall** provide a "Delete API Key" command that removes the stored credential from `SecretStorage` and confirms the deletion to the user.
REQ-53	Must	**The extension shall not** store API keys in `globalState`, `workspaceState`, or standard JSON settings files.
REQ-54	Must	**The extension shall** support dynamic Header Injection, populating authentication headers (e.g., `Authorization`, `X-Api-Key`) from `SecretStorage` at request time.
REQ-55	Should	**The extension shall** provide a first-run "Set API Key" command that writes the key to `SecretStorage` and confirms success to the user.

Model Context Protocol — Agentic Capabilities (§7.3)

ID	Priority	Requirement (EARS Format)
REQ-56	Must	**The extension shall** register as an MCP Server using the `vscode.lm.registerMcpServerDefinitionProvider` API.
REQ-57	Must	**The extension shall** expose extension-specific capabilities (e.g., `search_codebase`, `run_unit_tests`, `read_file_context`) as MCP Tools with JSON Schema parameter definitions.
REQ-58	Must	**The extension shall** act as the execution layer ("the hands"), receiving tool-call requests from the cloud-hosted model and executing them via the VS Code Extension API.
REQ-59	Must	**The extension shall** validate and sandbox MCP tool executions to prevent unintended side effects on the user's workspace.
REQ-59a	Must	**The extension shall** classify MCP tools by risk level (`safe`, `read-only`, `write-only`, `destructive`) and maintain an allowlist of external MCP hosts authorized to invoke each tool category.

ID	Priority	Requirement (EARS Format)
REQ-59b	Must	**When** the workspace is untrusted (per VS Code Workspace Trust API), **the extension shall** disable all tool execution except read-only context retrieval from open files.

Enterprise Graduation (§7.4)

ID	Priority	Requirement (EARS Format)
REQ-60	Must	**When** `BASE_URL` points to an enterprise gateway, **the extension shall** support SSO via the `vscode.authentication` provider, exchanging a VS Code session token for a gateway bearer token.
REQ-61	Must	**The extension shall** include an `X-Request-Id` header (UUID v4) in all outgoing LLM calls to enable end-to-end request tracing.
REQ-62	Must	**The extension shall** support a zero-code-change transition from direct SaaS access (BYOK) to enterprise-governed access (AI Gateway + SSO).
REQ-63	Must	**The extension shall** respect VS Code's built-in proxy settings and `HTTP_PROXY` / `HTTPS_PROXY` environment variables for corporate network environments.

Future-Proofing — Architectural Hedges (§7.6)

ID	Priority	Requirement (EARS Format)
REQ-64	Must	**The extension shall** encapsulate tool-call orchestration behind a `ToolOrchestrator` interface, allowing the iterative tool-call loop to be replaced with an alternative orchestration backend without modifying other extension components.
REQ-65	Must	**The extension shall** assign a `thread_id` (UUID v4) to every conversation or multi-turn interaction, available in-memory for the duration of the interaction.
REQ-66	Must	**The extension's** internal streaming layer **shall** emit typed events (minimally: `token`, `tool_call`, `tool_result`, `status`, `error`) rather than raw token strings, to support future event types without breaking the streaming contract.
REQ-67	Should	**The extension's** UI components for AI interactions **shall** support displaying step-by-step progress for multi-turn tool-call sequences, not assuming a single model response per user action.

Extension Lifecycle

ID	Priority	Requirement (EARS Format)
REQ-89	Must	**When** the extension is deactivated or reloaded, **the extension shall** abort in-progress requests, close open HTTP connections, and dispose of event listeners to prevent memory leaks.
REQ-90	Should	**The extension shall** implement graceful degradation during IDE shutdown: streaming requests shall be cancelled with a timeout of 5 seconds before forceful termination.
REQ-99	Must	**The extension shall** activate on the following events: (1) `onStartupFinished` for background MCP Server registration, (2) `onCommand:ourProduct.ai.*` for user-invoked AI commands, (3) `onView:ourProductAiChat` if a dedicated AI view is contributed. The extension shall complete activation within 2 seconds (per NFR).

URL Path Construction & Provider Type Behaviors (§8)

ID	Priority	Requirement (EARS Format)
REQ-68	Must	**The extension shall** treat the configured `baseUrl` as the complete base path to which API-specific paths (e.g., `/chat/completions`) are appended. For Azure OpenAI, users must include the full deployment path in `baseUrl`.
REQ-69	Must	**The extension shall** apply provider-specific behaviors (auth header name, default headers) based on the configured `type` field as specified in the Provider Type Behaviors table (§8).
REQ-70	Must	**The extension shall** allow users to override the default auth header name via the `authHeaderName` configuration field.
REQ-71	Must	**The extension shall** support three authentication modes via the `authMode` field: `"byok"` (SecretStorage), `"sso"` (vscode.authentication), and `"none"` (no authentication).

Tool Execution Safety (§7.3)

ID	Priority	Requirement (EARS Format)
REQ-72	Must	**The extension's** tool-call orchestration loop **shall** terminate after a configurable maximum number of iterations (default: 25) to prevent infinite loops.
REQ-73	Must	

ID	Priority	Requirement (EARS Format)
		The extension shall enforce a configurable timeout (default: 30 seconds) for individual tool executions to prevent indefinite hangs.
REQ-74	Must	**When** a tool is classified as destructive (e.g., `run_terminal_command`), **the extension shall** require explicit user approval before execution if `requireApprovalForDestructiveTools` is enabled.
REQ-75	Must	**The extension's** internal tool-call orchestrator **shall** invoke tools via direct function calls (not via the MCP protocol) to avoid serialization overhead for its own LLM interactions.

User Interaction & Safety

ID	Priority	Requirement (EARS Format)
REQ-76	Must	**The extension shall** support user-initiated cancellation of in-progress streaming requests via standard UI affordances (e.g., Escape key, Stop button).
REQ-76a	Should	**When** a streaming request is cancelled, **the extension shall** abort the HTTP request, clear partial UI state, and display a confirmation message (e.g., "Request cancelled").
REQ-76b	Should	**When** a request is in progress, **the extension shall** display a cancellable progress indicator with elapsed time.
REQ-77	Must	**When** a tool execution returns a result exceeding 10,000 characters, **the extension shall** truncate the result and append a notice (e.g., "[Output truncated; chars omitted]") before sending it to the model.
REQ-78	Must	**The extension shall** sanitize provider error messages before displaying to users or writing to logs, removing internal URLs, resource identifiers, and any string matching common API key patterns.

Priority Legend (MoSCoW)

- **Must:** Critical for launch; extension is not viable without it
- **Should:** Important but not critical; can launch without if necessary
- **Could:** Desirable; implement if time permits
- **Won't (v1):** Explicitly excluded from this version

11. Configuration Schema

VS Code Settings (`settings.json`)

Configuration uses VS Code's native settings system. Our extension contributes settings under a namespace (e.g., `ourProduct.ai`). This approach aligns with VS Code best practices and allows familiar configuration patterns.

Example: User or Workspace `settings.json`

```
{
  // AI Provider Configuration
  "ourProduct.ai.provider": {
    // Provider endpoint (required)
    "baseUrl": "https://mycompany.openai.azure.com/openai/
deployments/gpt-4",

    // Provider type for provider-specific behaviors (see §8
Provider Type Behaviors table)
    "type": "azure-openai",  // "azure-openai" | "openai" |
"anthropic" | "ollama" | "openshift-ai" | "bedrock" | "compatible"

    // Authentication mode: "byok" (Bring Your Own Key via
SecretStorage), "sso" (Enterprise SSO via vscode.authentication),
or "none" (no auth)
    "authMode": "byok",  // "byok" | "sso" | "none"

    // Override auth header name (optional; defaults per provider
type in §8 table)
    "authHeaderName": "Authorization",  // e.g., "Authorization",
"X-Api-Key", "api-key"

    // Additional headers (e.g., Azure API version)
    "headers": {
      "api-version": "2024-02-15-preview"
    },

    // Default model (can be overridden per feature)
    "defaultModel": "gpt-4-turbo"
  },

  // Model aliases - map friendly names to provider model IDs
  "ourProduct.ai.models": {
    "default": "gpt-4-turbo",
    "fast": "gpt-35-turbo",
    "reasoning": "gpt-4-turbo"
  },

  // Credential source (API key retrieved separately for security)
  "ourProduct.ai.credentials": {
    // Where to get the API key
    "source": "keychain",  // "keychain" | "environment" |
"prompt"
```

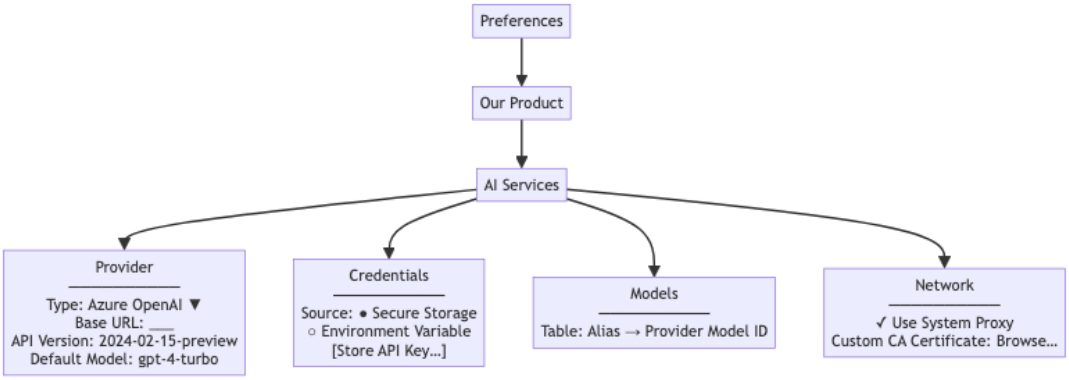
```
    // Environment variable name (if source is "environment")
    "environmentVariable": "OPENAI_API_KEY",

    // Keychain/Credential Manager entry name (if source is
"keychain")
    "keychainEntry": "ourProduct-ai-apikey"
  },

  // Feature-specific settings
  "ourProduct.ai.codeCompletion": {
    "enabled": true,
    "model": "fast",  // Uses model alias
    "maxTokens": 500
  },

  "ourProduct.ai.chat": {
    "enabled": true,
    "model": "default",
    "maxTokens": 4096,
    "temperature": 0.7
  },

  // Tool execution safety limits
  "ourProduct.ai.tooling": {
    // Maximum iterations for tool-call orchestration loop
(circuit breaker)
    "maxLoopIterations": 25,

    // Tool execution timeout in seconds
    "toolTimeout": 30,

    // Require user approval for destructive tools
(run_terminal_command, etc.)
    "requireApprovalForDestructiveTools": true
  },

  // Network settings
  "ourProduct.ai.network": {
    // Use VS Code's proxy settings, or override here
    "useSystemProxy": true,

    // Custom CA certificate for enterprise PKI
    "caCertificatePath": "/path/to/corporate-ca.crt",

    // Request timeout in seconds
    "timeout": 60
  },

  // Privacy settings
  "ourProduct.ai.telemetry": {
    "enabled": false  // Opt-in only
  }
}
```

Eclipse Preferences

Eclipse uses its preference system. Configuration is stored in workspace or user preferences. The extension shall provide a preference page under the product namespace accessible via **Window > Preferences > [Product Name] > AI Services**.

Implementation Requirements: - Preference page shall use Eclipse JFace preference framework with `FieldEditorPreferencePage` - Provider type shall be implemented as `ComboFieldEditor` with enum validation - API keys shall use `StringFieldEditor` with `echo` char set to `'*'` for masking - Base URL shall use `StringFieldEditor` with URL validation - Credentials shall be stored in Eclipse Secure Storage (`ISecurePreferences` API) under node `com.ourproduct.ai/credentials` - Preference changes shall trigger `IPropertyChangeListener` notifications to refresh active connections

Example: Preference Page Structure



Configuration Examples by Provider

Azure OpenAI

```
{
  "ourProduct.ai.provider": {
    "type": "azure-openai",
    "baseUrl": "https://{resource-name}.openai.azure.com/openai/
deployments/{deployment-name}",
    "authMode": "byok",
    "headers": {
      "api-version": "2024-02-15-preview"
    }
  }
}
```

Note: The extension appends `/chat/completions` to the `baseUrl`, so the full request URL becomes:

```
https://{resource-name}.openai.azure.com/openai/deployments/
{deployment-name}/chat/completions?api-version=2024-02-15-preview
```

OpenAI (Direct)

```
{
  "ourProduct.ai.provider": {
    "type": "openai",
    "baseUrl": "https://api.openai.com/v1",
    "defaultModel": "gpt-4-turbo"
  }
}
```

Enterprise Gateway

```
{
  "ourProduct.ai.provider": {
    "type": "compatible",
    "baseUrl": "https://ai-gateway.internal.mycompany.com/v1",
    "authMode": "sso",  // Use enterprise SSO instead of BYOK
    "headers": {
      "X-Department": "engineering",
      "X-Project-ID": "project-123"
    }
  }
}
```

Ollama (Self-Hosted)

```
{
  "ourProduct.ai.provider": {
    "type": "ollama",
    "baseUrl": "http://localhost:11434/v1",
    "defaultModel": "llama3"
  }
}
```

Note: For remote Ollama instances, use the appropriate hostname/IP (e.g., `http://ollama-server.internal:11434/v1`). No API key required for local instances; authentication can be configured if Ollama is secured.

Credential Storage Options

Method	Platform	How It Works
macOS Keychain	macOS	Stored in system Keychain; accessed via native APIs
Windows Credential Manager	Windows	Stored in Windows Credential Manager
Linux Secret Service	Linux	Stored via libsecret (GNOME Keyring, KWallet)
Environment Variable	All	Read from specified env var at runtime

Method	Platform	How It Works
Interactive Prompt	All	User prompted on first use; stored in credential store

Managed Configuration (v1.1 - Future)

For enterprise deployment, IT can distribute configuration profiles:

```
{
  // IT deploys this to all machines via GPO/MDM/config management
  "ourProduct.ai.managedConfig": {
    // Central configuration profile
    "profileUrl": "https://config.mycompany.com/ai-settings.json",

    // Settings that users cannot override
    "lockedSettings": [
      "ourProduct.ai.provider.baseUrl",
      "ourProduct.ai.telemetry.enabled"
    ],

    // Settings users can customize
    "userOverridable": [
      "ourProduct.ai.codeCompletion.enabled",
      "ourProduct.ai.chat.temperature"
    ]
  }
}
```

Configuration Validation

The extension validates configuration and provides helpful feedback:

Issue	User Feedback
Missing `baseUrl`	"AI provider not configured. Open Settings to configure."
Invalid URL format	"Invalid provider URL. Expected format: https://…"
Missing API key	"API key not found. Run 'Configure AI API Key' command."
Connection failed	"Cannot connect to AI provider. Check URL and network settings."
Authentication failed	"API key rejected by provider. Verify your API key is correct."

12. Error Handling

Provider Error Responses

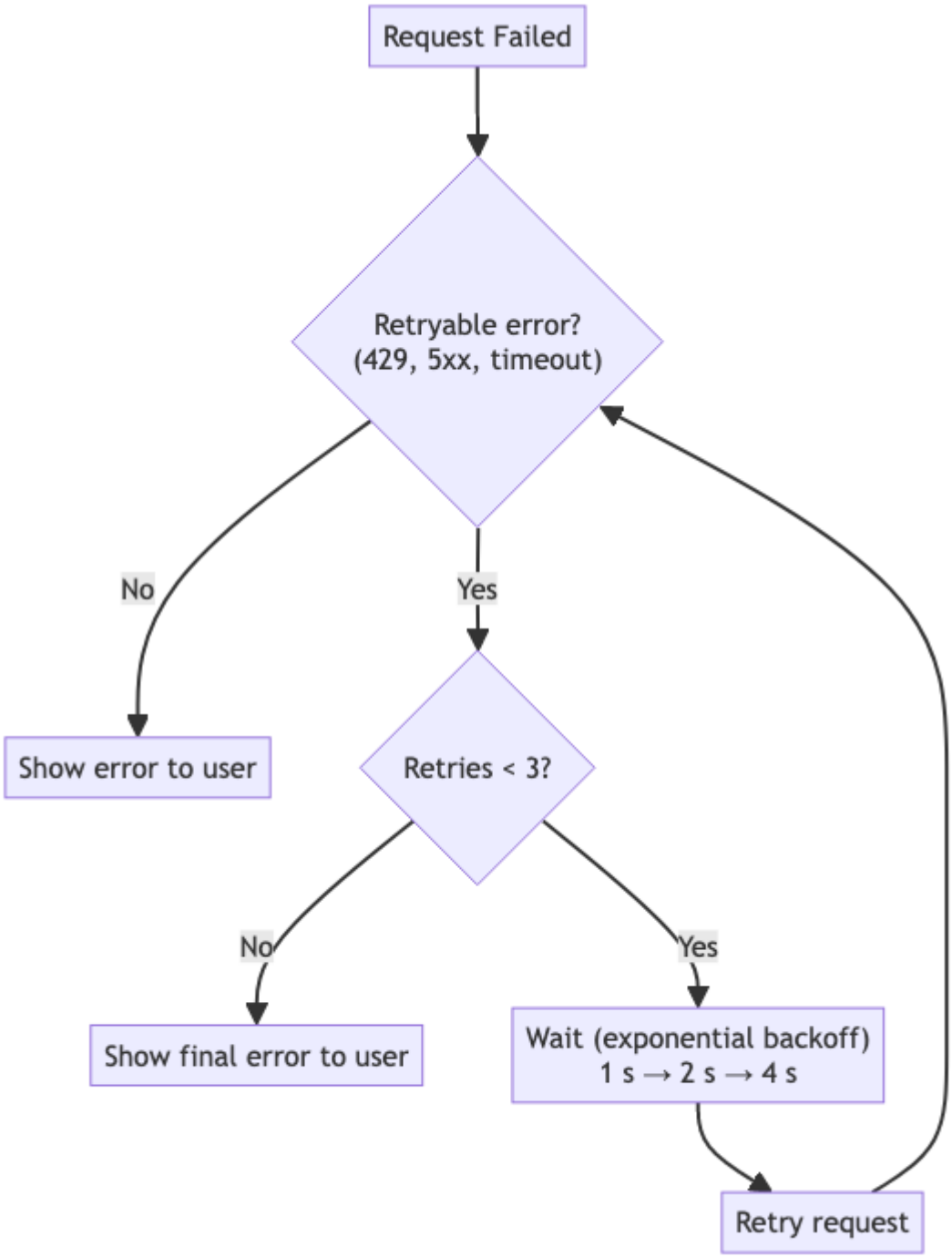
Provider Error	Extension Behavior	User Feedback
401 Unauthorized	No retry	"Authentication failed. Check your API key." + link to settings
403 Forbidden	No retry	"Access denied. Your API key may lack required permissions."
429 Rate Limited	Retry with backoff (respects `Retry-After`)	"Rate limited. Retrying…" or "Rate limit exceeded. Try again later."
400 Bad Request	No retry	"Request error: {sanitized provider message}" (see REQ-78 for sanitization rules)
500 Internal Error	Retry up to 3x with exponential backoff	"Provider error. Retrying…" or "Provider unavailable."
502/503 Unavailable	Retry up to 3x	"Provider temporarily unavailable. Retrying…"
Timeout	Retry once	"Request timed out. Retrying…" or "Request timed out. Check network."
Network Error	Check connectivity	"Cannot reach AI provider. Check your network connection."

Configuration Errors

Error	Detection	User Feedback
No provider configured	On activation	"AI not configured. Click here to set up."
Invalid base URL	On first request	"Invalid provider URL in settings."
Missing API key	On first request	"API key not found. Run 'Set API Key' command."
Credential store unavailable	On key retrieval	"Cannot access credential store. Check system keychain." (If

Error	Detection	User Feedback
		SecretStorage unavailable, fallback per REQ-07)

Retry Strategy



13. User Acceptance Testing (UAT) Plan

The following test cases must be passed to meet the **Definition of Done (DoD)**.

Test Group 1: Extension Installation & Activation

Test ID	Scenario	Expected Result	Priority
UAT-1.1	Install extension on VS Code (Windows)	Extension installs and activates without errors	Must
UAT-1.2	Install extension on VS Code (macOS)	Extension installs and activates without errors	Must
UAT-1.3	Install extension on VS Code (Linux)	Extension installs and activates without errors	Must
UAT-1.4	Install plugin in Eclipse	Plugin installs and activates without errors	Must
UAT-1.5	Extension activation with no configuration	Extension activates; shows “configure AI” prompt	Must

Test Group 2: Configuration

Test ID	Scenario	Expected Result	Priority
UAT-2.1	Configure via VS Code settings UI	Settings saved; AI features become available	Must
UAT-2.2	Configure via `settings.json` directly	Settings parsed correctly; AI features work	Must
UAT-2.3	Workspace settings override user settings	Workspace `baseUrl` used over user `baseUrl`	Must
UAT-2.4	Invalid `baseUrl` format	Validation error shown; setting not accepted	Must
UAT-2.5	Model alias resolves to provider model	Request uses mapped `provider_id`	Must

Test Group 3: Credential Management

Test ID	Scenario	Expected Result	Priority
UAT-3.1	Store API key in macOS Keychain	Key saved; retrieved on next use	Must
UAT-3.2	Store API key in Windows Credential Manager	Key saved; retrieved on next use	Must
UAT-3.3	Store API key via Linux Secret Service	Key saved; retrieved on next use	Must
UAT-3.4	Retrieve API key from environment variable	`OPENAI_API_KEY` env var used successfully	Must
UAT-3.5	No API key configured	User prompted to enter key; clear instructions shown	Must
UAT-3.6	Invalid API key	Clear error message; prompt to re-enter key	Must

Test Group 4: Provider Connectivity

Test ID	Scenario	Expected Result	Priority
UAT-4.1	Connect to Azure OpenAI	AI request succeeds with correct `api-version` header	Must
UAT-4.2	Connect to OpenAI API	AI request succeeds	Must
UAT-4.3	Connect via enterprise gateway	AI request succeeds with custom headers	Should
UAT-4.4	"Test Connection" command	Shows success/failure with helpful message	Should
UAT-4.5	Connection via corporate proxy	Request routed through configured proxy	Should

Test Group 5: AI Features

Test ID	Scenario	Expected Result	Priority
UAT-5.1	Code completion (text generation)	Streaming response displayed in editor	Must
UAT-5.2	Chat conversation	Multi-turn conversation works	Must

Test ID	Scenario	Expected Result	Priority
UAT-5.3	Function calling with `tools` parameter	Tool calls returned and handled	Must
UAT-5.4	Structured output (`response_format: json`)	Valid JSON response received	Must
UAT-5.5	Image generation (basic)	Image generated and displayed	Should
UAT-5.6	Streaming progress indicator	UI shows streaming in progress	Must

Test Group 6: Error Handling

Test ID	Scenario	Expected Result	Priority
UAT-6.1	Provider returns 401 (unauthorized)	Clear "check API key" message	Must
UAT-6.2	Provider returns 429 (rate limited)	Automatic retry; success or clear message	Must
UAT-6.3	Provider returns 500 (server error)	Retry with backoff; eventual error message	Must
UAT-6.4	Network timeout	Clear timeout message with troubleshooting hint	Must
UAT-6.5	No network connectivity	"Check network connection" message	Must

Test Group 7: Privacy & Security

Test ID	Scenario	Expected Result	Priority
UAT-7.1	Verify no telemetry without opt-in	Network capture shows no telemetry calls	Must
UAT-7.2	API key not in logs	Extension logs contain no API keys	Must
UAT-7.3	API key not in settings file (if keychain used)	`settings.json` contains no API key	Must
UAT-7.4	Prompts not persisted to disk	No prompt content in extension storage	Must

Test Group 8: Prompt Construction & Message Roles

Test ID	Scenario	Expected Result	Priority
UAT-8.1	Construct messages with system + user roles	Request sent with correct message array	Must
UAT-8.2	Multi-turn with assistant role	Conversation context preserved correctly	Must
UAT-8.3	Token count estimation	Estimate within 10% of actual token usage	Should
UAT-8.4	Context truncation with priority	High-priority context retained; low-priority truncated	Should
UAT-8.5	Prompt template substitution	Variables correctly replaced in template	Should

Test Group 9: Structured Output

Test ID	Scenario	Expected Result	Priority
UAT-9.1	Request with `response_format: json_object`	Valid JSON response received	Must
UAT-9.2	JSON response parsed to typed object	Response correctly typed; properties accessible	Must
UAT-9.3	Malformed JSON from provider	Graceful error with raw response for debugging	Must
UAT-9.4	Schema validation failure	Clear error indicating which field failed validation	Should

Test Group 10: Function Calling

Test ID	Scenario	Expected Result	Priority
UAT-10.1	Define and send tools array	Tools included in request; accepted by provider	Must
UAT-10.2	Receive and parse tool_calls	Function name and arguments correctly parsed	Must
UAT-10.3	Send tool result back to model	Tool role message accepted; conversation continues	Must
UAT-10.4	Multi-step tool calling loop	Multiple tool calls handled until final response	Should
UAT-10.5	`tool_choice: required`	Model returns tool_call, not text response	Should

Test ID	Scenario	Expected Result	Priority
	forces function call		
UAT-10.6	Parallel tool calls	Multiple simultaneous tool_calls handled correctly	Should

Test Group 11: Provider-Agnostic Interface (§7.1)

Test ID	Scenario	Expected Result	Priority
UAT-11.1	Configure custom `BASE_URL` in Settings	Extension sends requests to configured endpoint	Must
UAT-11.2	Switch `BASE_URL` from OpenAI to enterprise gateway	Extension calls new endpoint with zero code changes	Must
UAT-11.3	Configure custom `MODEL_NAME`	Requests include configured model identifier	Must
UAT-11.4	No vendor-specific SDK imported	Extension source does not depend on `@openai/openai` or `@anthropic-ai/sdk`	Must
UAT-11.5	Custom headers passed through	Provider receives custom headers (e.g., `api-version`) as configured	Must

Test Group 12: BYOK / SecretStorage (§7.2)

Test ID	Scenario	Expected Result	Priority
UAT-12.1	Store API key via "Set API Key" command	Key persisted in `SecretStorage`; confirmation shown	Must
UAT-12.2	API key injected into `Authorization` header	Outgoing request includes `Authorization: Bearer <key>`	Must
UAT-12.3	API key injected into `X-Api-Key` header	Outgoing request includes custom header with key from `SecretStorage`	Must
UAT-12.4	Key not present in `globalState` or settings	No API key found in `settings.json`, `globalState`, or `workspaceState`	Must
UAT-12.5			Should

Test ID	Scenario	Expected Result	Priority
	Key rotation via "Set API Key"	New key overwrites old; next request uses new key	

Test Group 13: MCP Agentic Capabilities (§7.3)

Test ID	Scenario	Expected Result	Priori
UAT-13.1	Extension registers as MCP Server	`vscode.lm.registerMcpServerDefinitionProvider` called on activation	Must
UAT-13.2	MCP tool `search_codebase` exposed	Tool listed with correct JSON Schema parameters	Must
UAT-13.3	MCP tool `read_file_context` executed	Returns file contents for requested path and line range	Must
UAT-13.4	Model tool-call triggers local execution	Extension executes tool via VS Code API and returns result	Must
UAT-13.5	Invalid tool call handled gracefully	Error returned to model without crashing extension	Shoul

Test Group 14: Enterprise Graduation (§7.4)

Test ID	Scenario	Expected Result	Priority
UAT-14.1	SSO login via `vscode.authentication`	Extension obtains bearer token from session; attaches to requests	Must
UAT-14.2	`X-Request-Id` header present on requests	All outgoing calls include a UUID v4 `X-Request-Id` header	Must
UAT-14.3	Proxy settings respected	Requests routed through VS Code / env var proxy configuration	Should
UAT-14.4	Switch from BYOK to SSO (zero code change)	Only `BASE_URL` and auth settings change; no extension code modified	Must

Test Group 15: Request Cancellation & Progress UX

Test ID	Scenario	Expected Result	Priority
UAT-15.1	Cancel streaming	Request aborted; UI cleared; "Request cancelled" message shown	Must

Test ID	Scenario	Expected Result	Priority
	request via Stop button		
UAT-15.2	Cancel via Escape key	Same behavior as Stop button	Should
UAT-15.3	Progress indicator during long request	Elapsed time shown; cancellable indicator visible	Should
UAT-15.4	Cancellation cleans up HTTP connection	No lingering connections after cancellation	Should

Test Group 16: Concurrent Requests & Multi-Provider

Test ID	Scenario	Expected Result	Priority
UAT-16.1	Open two chat sessions simultaneously	Both sessions work independently without blocking	Must
UAT-16.2	Rapid-fire multiple requests	Requests queued; rate limits respected	Should
UAT-16.3	Configure multiple provider profiles	Profiles saved in settings correctly	Should
UAT-16.4	Switch active provider profile	New profile applied; subsequent requests use new provider	Should

Test Group 17: Context Window Overflow & Provider Capabilities

Test ID	Scenario	Expected Result	Priority
UAT-17.1	Prompt exceeds 90% of context window	Warning shown before sending request	Should
UAT-17.2	Provider returns `context_length_exceeded`	Automatic truncation attempted; retry succeeds	Should
UAT-17.3	Truncation notification shown to user	User informed of context reduction; log shows removed messages	Should
UAT-17.4	Provider lacks vision support		Should

Test ID	Scenario	Expected Result	Priority
		Feature disabled gracefully; user notified	
UAT-17.5	Provider capability detection cached	`/v1/models` called once per session, not per request	Should

Test Group 18: Extension Lifecycle & Cleanup

Test ID	Scenario	Expected Result	Priority
UAT-18.1	Reload extension during active request	Request cancelled; connections closed cleanly	Must
UAT-18.2	Deactivate extension	All event listeners disposed; no memory leaks	Must
UAT-18.3	IDE shutdown with active request	Request cancelled within 5s timeout	Should
UAT-18.4	Configuration schema migration (upgrade)	Old settings automatically migrated to new schema	Should

Test Group 19: Provider-Specific Compatibility

Test ID	Scenario	Expected Result	Priority
UAT-19.1	Configure Ollama provider (type: ollama, authMode: none)	Connection succeeds; no auth header sent	Must
UAT-19.2	Ollama streaming response	Tokens stream correctly; UI updates incrementally	Must
UAT-19.3	Anthropic provider via OpenAI-compatible endpoint	Chat completions succeed; tool calling works	Must
UAT-19.4	OpenShift AI provider (KServe/vLLM backend)	Model list retrieved; completions succeed	Should
UAT-19.5	AWS Bedrock via OpenAI-	Requests routed correctly; responses parsed	Should

Test ID	Scenario	Expected Result	Priority
	compatible gateway		

Test Group 20: Security Edge Cases

Test ID	Scenario	Expected Result	Priority
UAT-20.1	Prompt injection via malicious HTML response	Response content-type validation fails; request rejected (REQ-46)	Must
UAT-20.2	SSO token expires during streaming request	Request fails gracefully; user prompted to re-authenticate	Must
UAT-20.3	SSO token refresh during active session	New token obtained transparently; request continues	Should
UAT-20.4	Custom CA certificate configured (corporate proxy)	HTTPS connection succeeds with corporate CA (REQ-43)	Must
UAT-20.5	CA certificate file missing or invalid	Clear error shown with CA setup instructions (REQ-45)	Should
UAT-20.6	API key deleted via “Delete API Key” command	Credential removed from SecretStorage; subsequent requests fail (REQ-52a)	Must
UAT-20.7	SecretStorage unavailable, OS stores unavailable	Persistent warning shown; env var fallback documented (REQ-07a)	Must
UAT-20.8	X-Request-Id header uniqueness	1000 concurrent requests all have unique UUIDs (REQ-58)	Should

Test Group 21: Tool Execution Safety

Test ID	Scenario	Expected Result	Priority
UAT-21.1	Tool-call loop exceeds maxLoopIterations (default 25)	Loop terminated; error returned to model (REQ-72)	Must
UAT-21.2	Individual tool execution exceeds	Tool cancelled; timeout error returned (REQ-73)	Must

Test ID	Scenario	Expected Result	Priority
	timeout (default 30s)		
UAT-21.3	Destructive tool requires approval (REQ-74)	User prompt shown before execution; requires confirmation	Must
UAT-21.4	Destructive tool in untrusted workspace	Tool execution blocked; read-only context only (REQ-59b)	Must
UAT-21.5	Tool result exceeds 10,000 characters	Result truncated with notice before sending to model (REQ-77)	Must

Test Group 22: Performance & NFR Validation

Test ID	Scenario	Expected Result	Priority
UAT-22.1	Measure extension activation time	< 2 seconds from activation trigger to ready state	Must
UAT-22.2	Measure extension memory footprint (idle)	< 100MB in IDE memory profiler	Must
UAT-22.3	Measure CPU usage (idle with no requests)	< 5% average CPU usage over 5 minutes	Should
UAT-22.4	Measure CPU usage (active streaming)	< 20% average during 60s streaming response	Should
UAT-22.5	Measure time to first token (after provider responds)	< 500ms from SSE first byte to UI display	Must
UAT-22.6	Measure extension overhead (request latency)	< 50ms overhead excluding network/provider time	Should
UAT-22.7	Extension bundle size (VS Code)	< 10MB .vsix package size	Should
UAT-22.8	Extension bundle size (Eclipse)	< 15MB with dependencies	Should
UAT-22.9	Network timeout:	Fails after 10 seconds (REQ-93)	Must

Test ID	Scenario	Expected Result	Priority
	connection establishment		
UAT-22.10	Network timeout: streaming read	Fails after 60 seconds inactivity (REQ-93)	Must
UAT-22.11	Network timeout: non-streaming read	Fails after 30 seconds (REQ-93)	Must
UAT-22.12	Response size limit enforcement	1MB response truncated with warning (REQ-94)	Must

Test Group 23: Accessibility Compliance

Test ID	Scenario	Expected Result	Priority
UAT-23.1	Keyboard-only navigation (Command Palette)	All AI commands accessible via keyboard (REQ-95)	Must
UAT-23.2	Screen reader announces streaming status	“Thinking…”, “Generating response…”, “Response complete” announced (REQ-96)	Must
UAT-23.3	Focus management during streaming	Focus predictable; status changes announced	Must
UAT-23.4	High contrast theme support	UI elements visible in high contrast mode	Should
UAT-23.5	Zoom level 200% (VS Code)	UI remains usable at 200% zoom	Should

Test Group 24: Telemetry & Observability

Test ID	Scenario	Expected Result	Priority
UAT-24.1	Telemetry disabled by default	No network calls to telemetry endpoints without opt-in (REQ-44)	Must
UAT-24.2	Telemetry opt-in via settings UI	Setting enabled; events collected (REQ-92)	Should
UAT-24.3	Telemetry event schema validation	Events match defined schema: ai.request.started, completed, failed, tool.executed (REQ-91)	Should

Test ID	Scenario	Expected Result	Priority
UAT-24.4	Telemetry excludes PII	No prompts, API keys, file paths in telemetry payloads (REQ-91)	Must
UAT-24.5	X-Request-Id in all LLM requests	UUID v4 header present in captured network traffic (REQ-58)	Must

Test Group 25: Internationalization Readiness

Test ID	Scenario	Expected Result	Priority
UAT-25.1	English (en-US) locale default	All UI strings display correctly in English (REQ-98)	Must
UAT-25.2	Code inspection: externalized strings	All user-facing strings use i18n framework (REQ-97)	Should
UAT-25.3	Message key convention validation	Keys follow `extension.category.message` pattern (REQ-98)	Should

Test Group 26: Architectural Hedge Validation (§7.6)

Test ID	Scenario	Expected Result	Priority
UAT-26.1	ToolOrchestrator interface abstraction	Tool-call loop implementation behind interface/abstract class (REQ-64)	Must
UAT-26.2	thread_id assignment on new conversations	Every conversation assigned UUID v4 thread_id (REQ-65)	Must
UAT-26.3	Event-typed streaming layer	Streaming emits typed events (token, tool_call, tool_result, status, error) not raw strings (REQ-66)	Must
UAT-26.4	Multi-step UI progress display	UI shows step-by-step tool-call progress, not single progress bar (REQ-67)	Should

---

14. Non-Functional Requirements (NFRs)

Category	Requirement	Target	Measurement
Latency		< 50ms	

Category	Requirement	Target	Measurement
	Extension overhead (excluding provider time)		Performance profiling
Responsiveness	Time to first token displayed	< 500ms after provider responds	User testing
Memory	Extension memory footprint	< 100MB	IDE memory profiling
Startup	Extension activation time	< 2 seconds	Activation timing
Compatibility	VS Code versions supported	Latest 3 major versions	CI test matrix
Compatibility	Eclipse versions supported	Latest 2 major versions	CI test matrix
Platforms	OS support	Windows 10+, macOS 11+, Ubuntu 20.04+	CI test matrix
Security	Credential exposure	Zero plaintext API keys in settings/logs	Security audit
Privacy	Telemetry	Opt-in only; no data without consent	Privacy review
Accessibility	Screen reader compatibility	WCAG 2.1 AA	Accessibility testing
Accessibility	Keyboard navigation	All AI commands accessible via keyboard shortcuts	Manual testing
Accessibility	Focus management	Focus moves predictably during streaming; status announced to screen readers	Accessibility testing
Compatibility	Configuration schema versioning	Automatic migration of deprecated settings	Unit tests
Compatibility	Backward compatibility	Support settings from previous 2	Integration tests

Category	Requirement	Target	Measurement
		major versions	
Bundle Size	Extension package size	< 10MB (VS Code), < 15MB (Eclipse with dependencies)	Build artifact size check
CPU	Idle CPU usage	< 5% when no AI requests active	Profiling
CPU	Active CPU usage	< 20% during streaming response	Profiling
i18n	Localization support	English (v1); extensible architecture for future locales (v1.1+)	Code review
Network	Connection timeout	10 seconds	Integration tests
Network	Read timeout (streaming)	60 seconds	Integration tests
Network	Read timeout (non-streaming)	30 seconds	Integration tests
Response Size	Maximum response size	1MB (truncate with notice if exceeded)	Unit tests
Telemetry	Event schema	Defined event types with anonymized data fields (see REQ-91)	Schema validation

15. Glossary

Term	Definition
IDE Extension	Our plugin for VS Code or Eclipse that provides AI-powered developer features
Provider	The AI cloud service (Azure OpenAI, OpenAI, enterprise gateway) that processes requests
OpenAI API	REST API specification for LLM interactions; industry standard implemented by multiple providers
Azure OpenAI	

Term	Definition
	Microsoft's Azure-hosted OpenAI service; common enterprise choice
Enterprise Gateway	Customer's internal API gateway that may proxy requests to AI providers
base_url	The endpoint URL for the AI provider (e.g., `https://api.openai.com/v1`)
Model Alias	User-friendly name (e.g., "fast") mapped to provider model ID (e.g., "gpt-3.5-turbo")
SSE	Server-Sent Events — HTTP streaming protocol for real-time token delivery
Keychain	OS-level secure credential storage (macOS Keychain, Windows Credential Manager)
settings.json	VS Code's JSON configuration file for user and workspace settings
Message Role	The role of a message in a conversation: `system`, `user`, `assistant`, or `tool`
System Prompt	Instructions defining the AI's behavior, persona, and constraints
Prompt Template	Reusable prompt structure with variable placeholders for programmatic use
Function Calling	OpenAI API feature allowing models to invoke defined functions/tools
Tools	Function definitions sent to the model describing callable operations
tool_calls	Model's response requesting execution of defined functions
Structured Output	OpenAI API feature for requesting JSON-formatted responses (`response_format`)
JSON Mode	Setting `response_format: { type: "json_object" }` to ensure valid JSON output
Few-shot Prompting	Including examples in the prompt to guide model output format
Context Window	Maximum tokens (input + output) a model can process in one request
Token	Basic unit of text processing; roughly 4 characters or 0.75 words in English
EARS	Easy Approach to Requirements Syntax — format for unambiguous requirements
MoSCoW	Prioritization framework: Must have, Should have, Could have, Won't have
MCP	Model Context Protocol — standard for AI tool integration; used for agentic capabilities via `vscode.lm` API (see §7.3)
MCP Server	An extension or service that exposes tools and capabilities via the Model Context Protocol

Term	Definition
MCP Client / Host	A consumer that discovers MCP tools and orchestrates their invocation; the extension acts as both MCP Server *and* a lightweight MCP Client for its own tool-call loop
MCP Tool	A capability exposed by an MCP Server (e.g., `search_codebase`, `read_file_context`) described with JSON Schema parameters
Tool-Call Orchestrator	The modular loop that sends tool definitions to the model, executes returned `tool_calls`, and repeats until a final text response — encapsulated behind a `ToolOrchestrator` interface for future replaceability
thread_id	UUID v4 assigned to each conversation or multi-turn interaction for correlation; maps to future agent framework thread concepts (e.g., LangGraph threads)
Event-Typed Streaming	Internal streaming architecture that emits typed events (`token`, `tool_call`, `tool_result`, `status`, `error`) rather than raw strings, enabling future extension without contract breakage
LangGraph	A Python-based agent framework by LangChain for stateful, graph-defined AI workflows; explicitly out of scope for v1 but architecture preserves future optionality (see §7.6)
Universal LLM Client	The extension's provider-agnostic HTTP client that targets the OpenAI Chat Completions wire format without vendor SDKs (see §7.1)
SecretStorage	VS Code API (`context.secrets`) for persisting sensitive data (e.g., API keys) in OS-level secure storage
Header Injection	Dynamic population of HTTP authentication headers from `SecretStorage` at request time
BYOK	Bring Your Own Key — users supply their own API key for direct SaaS access during evaluation
Gateway Graduation	Zero-code-change transition from direct SaaS (BYOK) to enterprise-governed AI Gateway access (see §7.4)
vscode.authentication	VS Code API for SSO / OIDC / OAuth authentication flows used in enterprise gateway integration
X-Request-Id	UUID v4 header included in all outgoing LLM calls for end-to-end observability and tracing
AI Gateway	Enterprise-managed API gateway (e.g., Kong, Cloudflare AI Gateway) that governs LLM traffic
Red Hat OpenShift AI	Kubernetes-native AI/ML platform (part of Red Hat OpenShift) that provides model serving via KServe/vLLM with OpenAI-compatible inference endpoints
Agents	Autonomous AI systems that can take actions; v1 supports model-driven tool-call loops via MCP (no

Term	Definition
	local agent runtime). Full agent frameworks (LangGraph, etc.) deferred to v2+ (see §6, §7.6)

16. Revision History

Version	Date	Author	Changes
0.1	2026-01-01	—	Initial draft
0.2	2026-01-07	—	Added: Problem Statement, Success Metrics, Assumptions, Out of Scope, Error Handling Matrix, Glossary. Fixed: diagram syntax, JSON formatting. Enhanced: MoSCoW prioritization, UAT coverage.
0.3	2026-01-07	—	Reframed as B2B multi-tenant SaaS. Added: Multi-Tenancy section, compliance alignment, tenant API.
0.4	2026-01-07	—	Reframed as on-premises with separate backend component.
0.5	2026-01-07	—	**Major revision: Client-side IDE extension model.** Removed backend/gateway—extension calls provider directly. Target: Windows/macOS/Linux workstations. Config via IDE-native settings. Credentials via OS keychain. In-scope: text/code generation, structured output, function calling, limited images. Out of scope: Agents, MCP, audio. Future: Central config management (v1.1).
0.6	2026-01-07	—	**Added SDK/API patterns for developers.** New §8 Prompt Engineering & API Patterns: message roles, prompt templates, instruction following, structured output handling, function calling (tools), and context management. Added 19 new requirements (REQ-28 to REQ-46) for SDK capabilities. Added 3 new UAT test groups: prompt construction, structured output, function calling. Expanded glossary with prompt engineering terms.
0.7	2026-02-26	—	**Design-phase requirements for VS Code extension.** New §7 VS Code Extension Design-Phase Requirements superseding prior guidance on provider integration, credential storage, and tool capabilities. Key additions: (1) Provider-Agnostic Interface — Universal LLM Client via OpenAI wire format, no vendor SDKs, configurable `BASE_URL`/`MODEL_NAME`. (2)

Version	Date	Author	Changes
			BYOK via VS Code `SecretStorage` API with header injection. (3) MCP moved **in-scope** — extension registers as MCP Server via `vscode.lm` API, exposes tools. (4) Enterprise Graduation — `vscode.authentication` SSO support, `X-Request-Id` tracing header. Added 17 new requirements (REQ-47 to REQ-63). Added 4 new UAT test groups (11–14). Updated Architecture Decisions table. Expanded glossary with MCP, SecretStorage, BYOK, Gateway Graduation, and related terms. Sections renumbered (8–17).
0.8	2026-02-26	—	**Clarified MCP architecture; added future-proofing hedges.** (1) §7.3 rewritten to clarify dual MCP role (Server + Tool-Call Orchestrator) — explicitly states no local agent runtime is required; cloud model drives the tool loop, extension is a stateless executor. (2) LangGraph / agent frameworks added to §6 Out of Scope with rationale. (3) New §7.6 Future-Proofing: four architectural hedges (modular `ToolOrchestrator` interface, `thread_id` correlation, event-typed streaming, multi-step UX readiness) to preserve agent-framework optionality at minimal v1 cost. (4) Added 4 new requirements (REQ-64 to REQ-67). (5) Expanded glossary with MCP Client/Host, Tool-Call Orchestrator, `thread_id`, Event-Typed Streaming, LangGraph.
0.9	2026-02-26	—	**Provider compatibility update.** Removed "Anthropic Claude Adapter" and "AWS Bedrock Adapter" from Deferred table — Anthropic's OpenAI-compatible endpoint eliminates the need for a translation layer; Anthropic Claude promoted to ✅ v1 supported provider. AWS Bedrock reclassified as conditionally supported via OpenAI-compatible endpoint or enterprise gateway; native SigV4 auth deferred to v1.1. Updated Supported Providers table, §7.5 routing row, and configuration type enum.
0.10	2026-02-26	—	**Added Red Hat OpenShift AI** to Supported Providers (v1) — OpenAI-compatible via KServe/vLLM model serving runtime. No adapter required. Added glossary entry.

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0.11	2026-02-26	—	**P0 Implementation Blockers Resolved.** (1) Clarified Azure OpenAI URL path construction: `baseUrl` is the complete base path including deployment; extension appends `/chat/completions`. (2) Added §8 Provider Type Behaviors table specifying auth header, path construction, and default headers per provider type (azure-openai, openai, anthropic, ollama, openshift-ai, bedrock, compatible). (3) Added `authMode` config field (`byok`, `sso`, `none`) to explicitly control authentication method. (4) Added `authHeaderName` config field to override default auth header per provider. (5) Clarified in §7.3 that internal tool-call orchestrator uses direct function calls (not MCP protocol) for performance. (6) Added tool execution safety limits: `maxLoopIterations` (default 25), `toolTimeout` (default 30s), `requireApprovalForDestructiveTools` flag. (7) Upgraded REQ-43 (CA certs), REQ-46 (prompt injection), REQ-59 (sandboxing), REQ-63 (proxy) from Should to Must for enterprise-critical capabilities. (8) Added 8 new requirements (REQ-68 to REQ-75) covering URL path construction, provider behaviors, auth modes, and tool safety. Updated provider type enum to include `openshift-ai` and `bedrock`.
0.12	2026-02-26	—	**P1 Security Critical Improvements.** (1) Added MCP tool permission model in §7.3: tools classified by risk level (`safe`, `read-only`, `write-only`, `destructive`); allowlist for external MCP hosts per tool category; destructive tools (e.g., `run_terminal_command`) internal-only by default. (2) Added Workspace Trust integration: tool execution disabled in untrusted workspaces; only read-only context from open files permitted. (3) Added error message sanitization (REQ-78): provider error messages sanitized before display to remove internal URLs, resource identifiers, API key patterns. (4) Updated `run_terminal_command` tool description with security warning and classification. (5) Reconciled REQ-07 (OS credential stores) with REQ-52 (SecretStorage): SecretStorage is primary, OS stores are

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			fallback when SecretStorage unavailable. (6) Added REQ-52a: “Delete API Key” command for credential removal. (7) Added REQ-76: user-initiated request cancellation support. (8) Added REQ-77: tool result truncation at 10,000 characters to prevent context window overflow. (9) Added REQ-59a (tool risk classification), REQ-59b (untrusted workspace restrictions). Updated error handling section to reference REQ-78 sanitization.
0.13	2026-02-26	—	**P2 Contradictions & Structural Fixes.** (1) Fixed REQ-06 settings precedence: now correctly specifies workspace settings take precedence over user settings, matching VS Code convention. (2) Fixed §7.5 table: MCP tools now shown as available in both evaluation and enterprise modes (was incorrectly shown as enterprise-only). (3) Added explicit “Supersedes” list in §7 documenting which pre-v0.7 requirements are superseded (REQ-07, REQ-10, REQ-14, REQ-17). (4) Added note explaining REQ-24 through REQ-27 numbering gap (removed during v0.6 refactoring, gap preserved for traceability). (5) Updated §4 Dependencies table: credential helper row now shows SecretStorage as Complete (v0.7) instead of stale “Not Started” status. (6) Removed orphaned “N/A (separate SRS)” reference for Agents/Assistants API; replaced with clear scoping: deferred indefinitely, requires separate architecture study. (7) Defined Image Generation “limited” scope: single image generation only; batch generation, image editing, and variations not supported in v1.
0.14	2026-02-26	—	**P3 Implementation Gaps Resolved.** (1) Enhanced request cancellation UX: added REQ-76a (cancellation cleanup and confirmation), REQ-76b (cancellable progress indicator with elapsed time). (2) Added concurrent request handling: REQ-79 (multiple concurrent chat sessions), REQ-80 (request queuing and rate limiting). (3) Added provider capability detection: REQ-81 (detect vision/multimodal support), REQ-82 (graceful degradation for unsupported features), REQ-83 (capability caching). (4) Added context window overflow handling:

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			REQ-86 (90% threshold warning), REQ-87 (automatic truncation and retry on token limit errors), REQ-88 (user notification of context reduction). (5) Added multi-provider configuration: REQ-84 (multiple named provider profiles), REQ-85 (runtime profile switching via UI command). (6) Added extension lifecycle management: REQ-89 (cleanup on deactivation/reload), REQ-90 (graceful shutdown with 5s timeout). (7) Added REQ-21a (progress indication for waiting users). (8) Enhanced Eclipse Preferences section with implementation requirements: `FieldEditorPreferencePage`, `ISecurePreferences`, `IPropertyChangeListener`, field editor specifications. (9) Added NFR rows for configuration schema versioning and backward compatibility (previous 2 major versions supported). (10) Added 4 new UAT test groups (15-18): Request Cancellation & Progress UX, Concurrent Requests & Multi-Provider, Context Window Overflow & Provider Capabilities, Extension Lifecycle & Cleanup — covering 16 new test scenarios.
0.15	2026-02-26	—	**P4 Non-Functional Requirements Detailed.** (1) Added 8 NFR table rows: Bundle Size (< 10MB VS Code, < 15MB Eclipse), CPU usage (< 5% idle, < 20% active), i18n support (English v1, extensible architecture), detailed accessibility (keyboard navigation, focus management, screen reader announcements), network timeouts (10s connection, 60s streaming, 30s non-streaming), response size limit (1MB truncation), telemetry event schema. (2) Added 9 new requirements: REQ-07a (complete credential storage failure handling with env var fallback), REQ-91 (telemetry event schema with anonymized metadata), REQ-92 (telemetry settings UI), REQ-93 (network timeout enforcement), REQ-94 (response size truncation at 1MB), REQ-95 (keyboard-only operation), REQ-96 (screen reader status announcements), REQ-97 (i18n framework with externalized strings), REQ-98 (English locale v1 with translation-ready

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			keys), REQ-99 (activation events: `onStartupFinished`, `onCommand`, `onView` with 2s activation limit). (3) Enhanced Security & Privacy section with new "Telemetry & Observability" subsection. (4) Enhanced User Experience section with new "Accessibility & Internationalization" subsection grouping a11y and i18n requirements.
0.16	2026-02-26	—	**P5 Test Coverage Gaps Resolved.** Added 8 comprehensive UAT test groups (19-26) covering all remaining test coverage gaps with 47 new test scenarios: (1) **Test Group 19: Provider-Specific Compatibility** — 5 tests for Ollama (authMode: none), Anthropic OpenAI-compatible endpoint, OpenShift AI (KServe/vLLM), AWS Bedrock gateway routing. (2) **Test Group 20: Security Edge Cases** — 8 tests for prompt injection mitigation (REQ-46), SSO token expiry/refresh, custom CA certificates (REQ-43, REQ-45), API key deletion (REQ-52a), complete credential storage failure (REQ-07a), X-Request-Id uniqueness (REQ-58). (3) **Test Group 21: Tool Execution Safety** — 5 tests for maxLoopIterations enforcement (REQ-72), tool timeout (REQ-73), destructive tool approval (REQ-74), untrusted workspace restrictions (REQ-59b), tool result truncation (REQ-77). (4) **Test Group 22: Performance & NFR Validation** — 12 tests measuring activation time, memory footprint, CPU usage (idle/active), time to first token, extension overhead, bundle sizes, network timeouts (REQ-93), response size limits (REQ-94). (5) **Test Group 23: Accessibility Compliance** — 5 tests for keyboard-only navigation (REQ-95), screen reader announcements (REQ-96), focus management, high contrast themes, 200% zoom support. (6) **Test Group 24: Telemetry & Observability** — 5 tests for opt-in enforcement (REQ-44), event schema validation (REQ-91), PII exclusion, X-Request-Id presence (REQ-58). (7) **Test Group 25: Internationalization Readiness** — 3 tests for English locale default (REQ-98), externalized strings (REQ-97), message key conventions. (8) **Test Group 26: Architectural Hedge**

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			Validation (§7.6) — 4 tests for ToolOrchestrator interface abstraction (REQ-64), thread_id assignment (REQ-65), event-typed streaming (REQ-66), multi-step UI progress (REQ-67). Comprehensive test coverage now ensures all 99 functional requirements and 15 NFRs are validated through structured UAT scenarios.

17. Appendix: Related Documentation

- OpenAI API Reference
- OpenAI Platform Concepts
- OpenAI Rate Limits
- Azure OpenAI Service Documentation
- vLLM Documentation
- Model Context Protocol Specification
- VS Code SecretStorage API
- VS Code Authentication API
- VS Code Language Model API